



Déploiement et paramétrage de PfSense

PfSense est un routeur/pare-feu open source basé sur le système d'exploitation FreeBSD. Il utilise le pare-feu à états Packet Filter, des fonctions de routage et de NAT lui permettant de connecter plusieurs réseaux informatiques. Le déployer dans un réseau local virtualisé permet d'avoir à disposition une solution de routeur/pare-feu très complète. L'idée de ce TP est de monter un routeur fonctionnel pour connecter un réseau local virtualisé à Internet

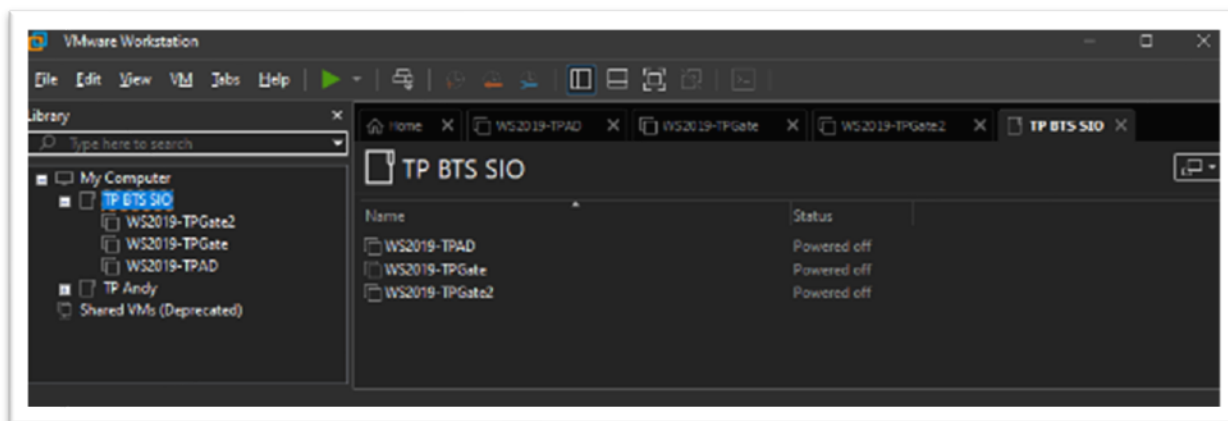
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Création d'une machine virtuelle sous VMWare Workstation

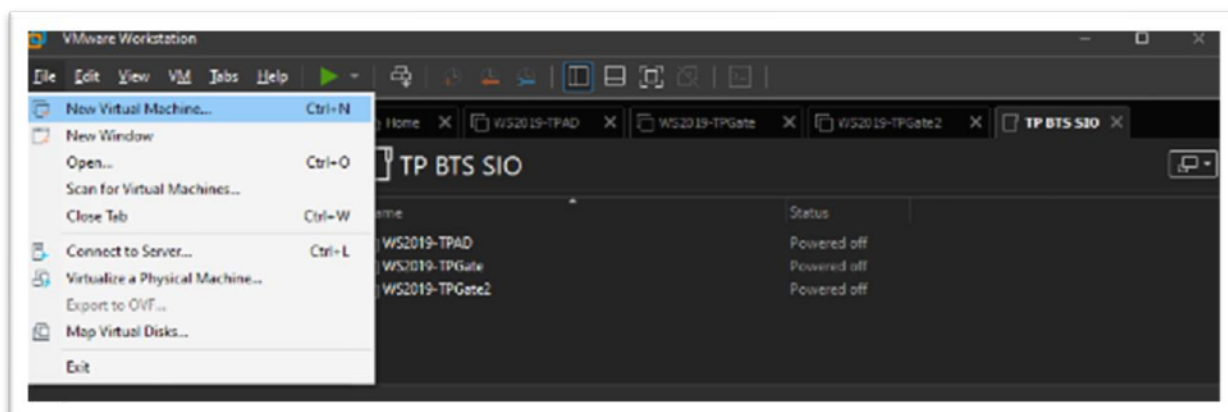
Créer une nouvelle machine virtuelle

Choisir un emplacement pour la nouvelle machine virtuelle. Ici, dans mon dossier "TP BTS SIO"



Menu "File"

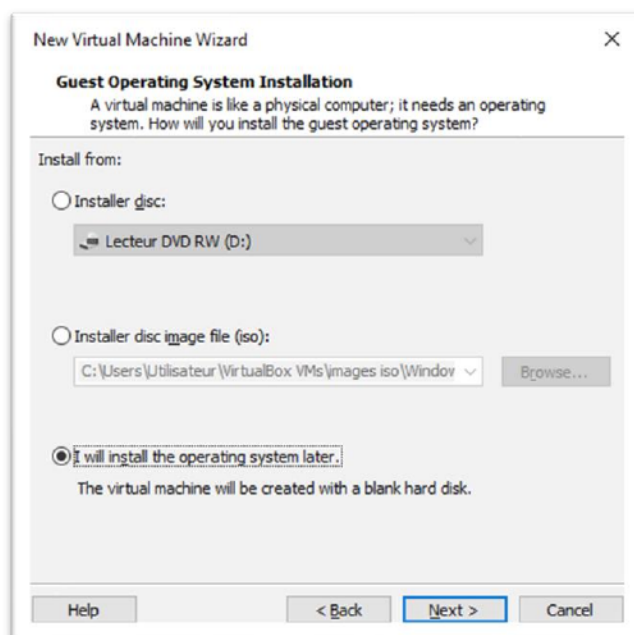
Sélectionner "New Virtual Machine"



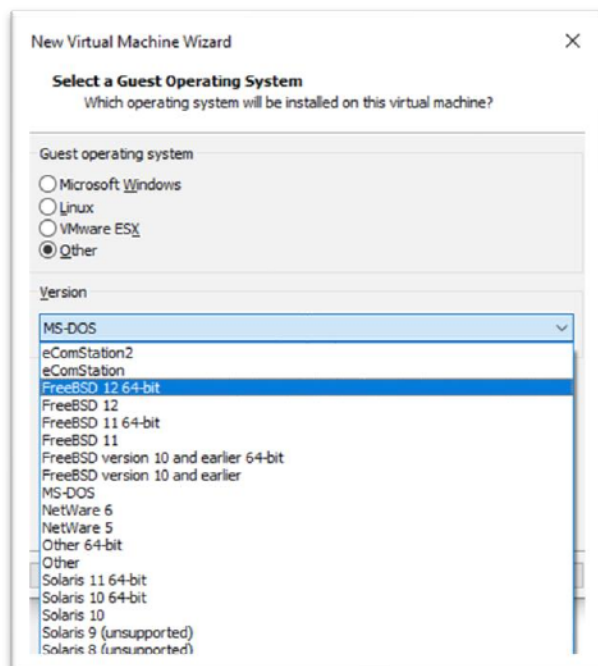
Sélectionner l'option "Typical"



Sélectionner "I will Install the operating system later"



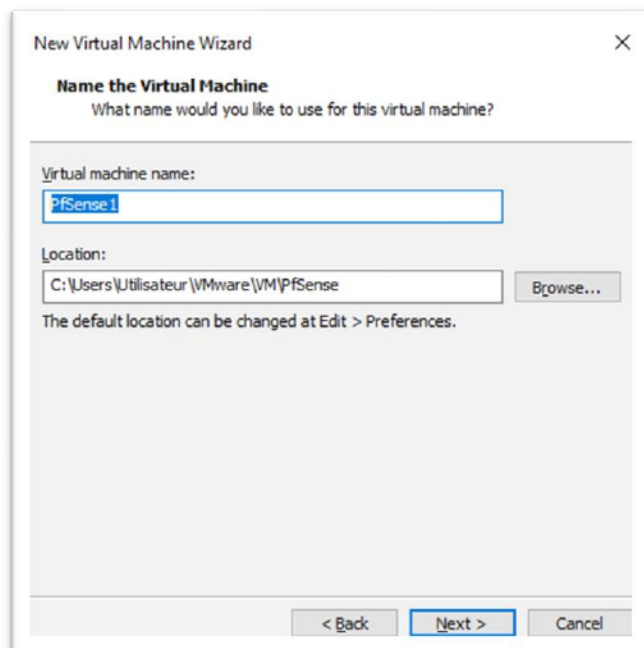
Choisir l'option "Other" et sélectionner la version FreeBSD qui convient à l'ordinateur sur lequel VMWare est installé. Ici "FreeBSD 12 64-bit"



FreeBSD est un système d'exploitation libre UNIX, développé par l'université de Berkeley, en Californie (BSD = Berkeley Software Distribution). Il est conçu pour faire fonctionner les plates-formes de type serveur, station de travail et les systèmes embarqués modernes. Une importante communauté l'a développé continuellement pendant plus de trente ans. Ses fonctionnalités réseau, de sécurité et de stockage avancées ont fait de FreeBSD la plate-forme choisie par certains des sites Web les plus visités ainsi que pour la plupart des systèmes embarqués orientés réseau et des systèmes de stockage les plus répandus.

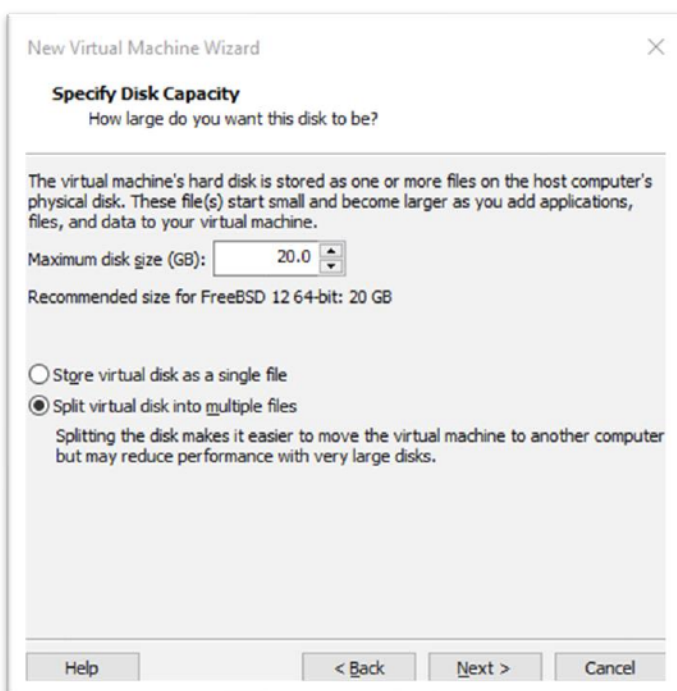
Donner un nom à la machine virtuelle et choisir son dossier d'installation.

Ici, PfSense1

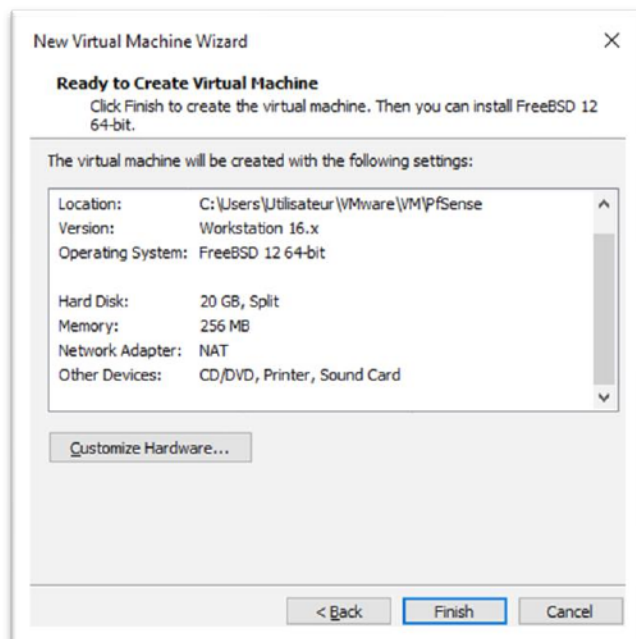


Taille du disque. Laisser 20 Go même si un Firewall-routeur n'a guère plus besoin de 1 Go. VMWare gère l'espace de stockage de façon dynamique jusqu'à la taille maximum que l'on attribue à la machine virtuelle. Ici 20 Go.

Choisir l'option "Split virtual disk into multiple files"

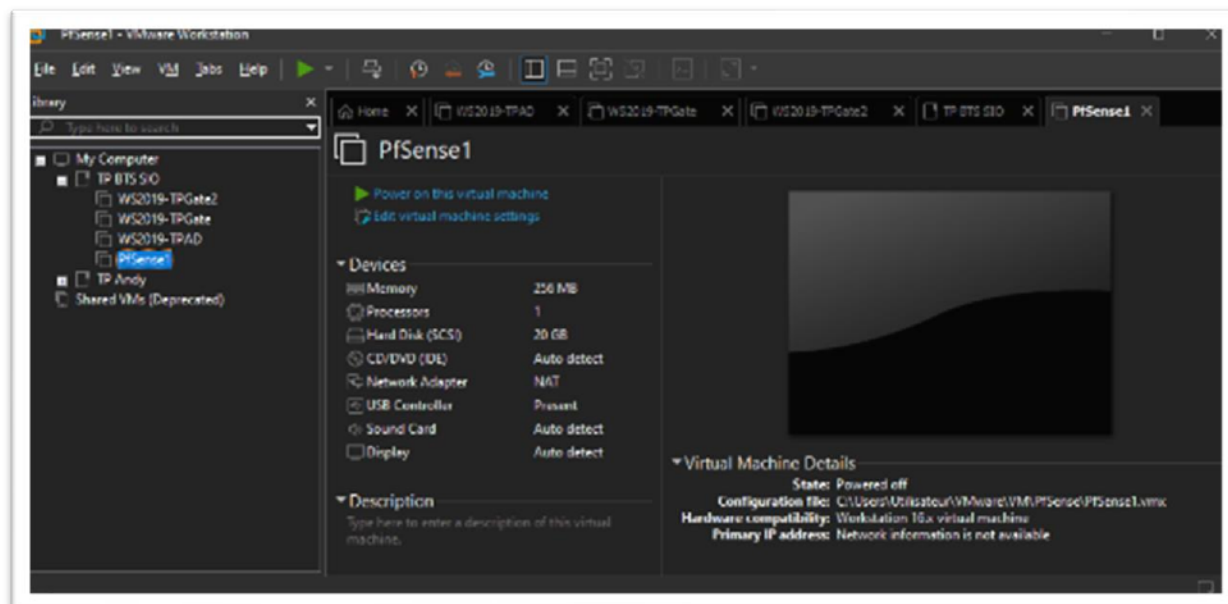


Cliquer sur Finish pour terminer l'installation



Configurer la nouvelle machine virtuelle

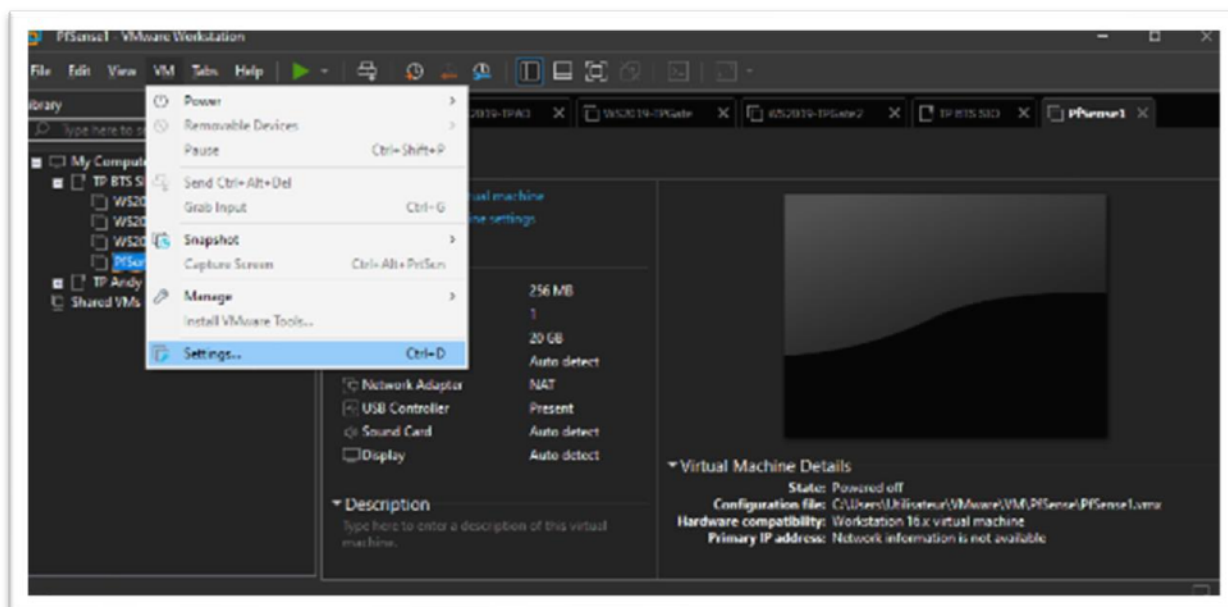
Replacer la machine virtuelle dans le bon dossier si nécessaire.



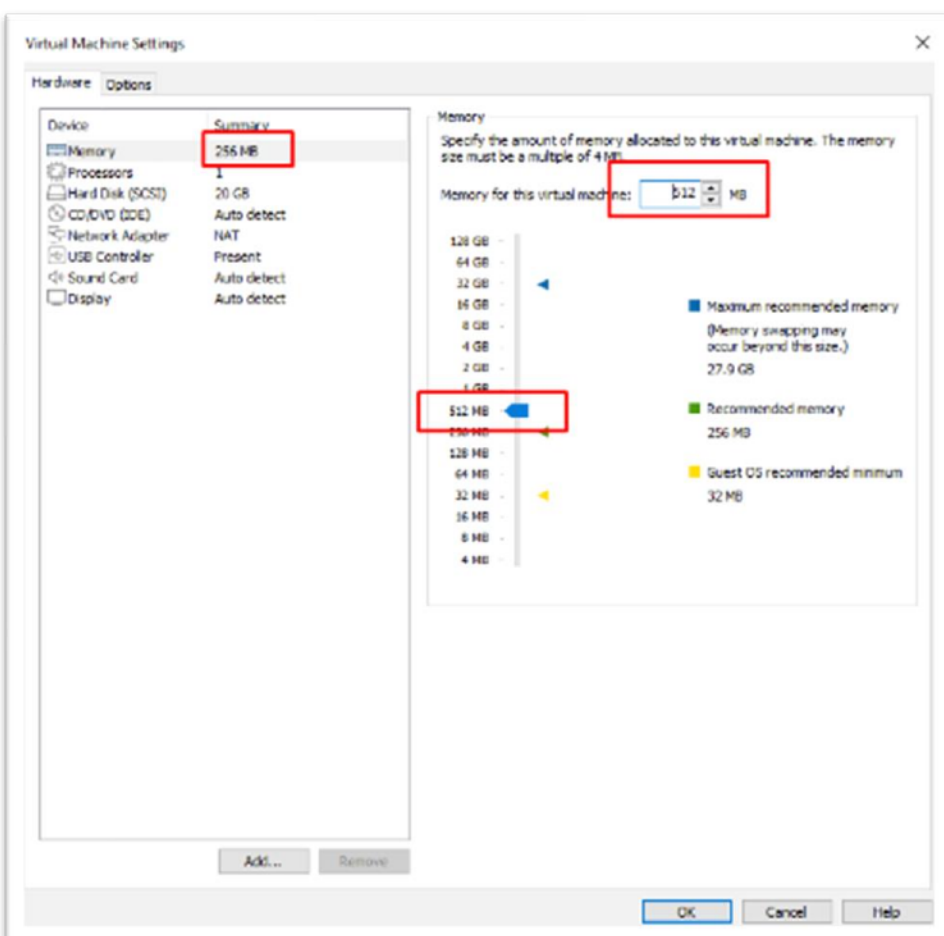
Sélectionner la machine virtuelle PfSense

Menu "VM"

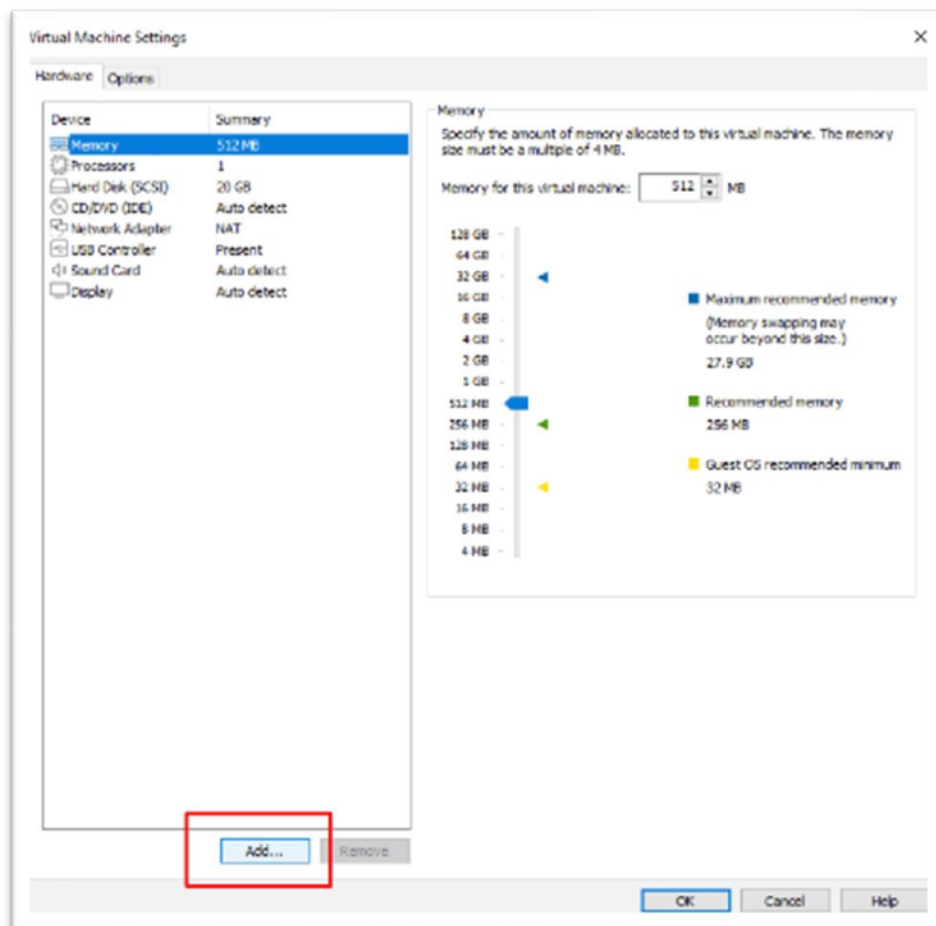
Sélectionner "Settings..."



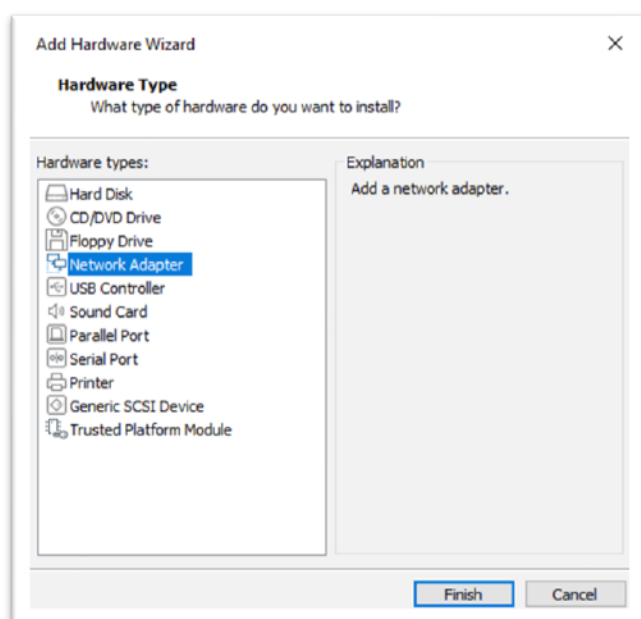
Dans l'onglet "Hardware", sélectionner "Memory" et allouer 512MB de RAM à la machine virtuelle



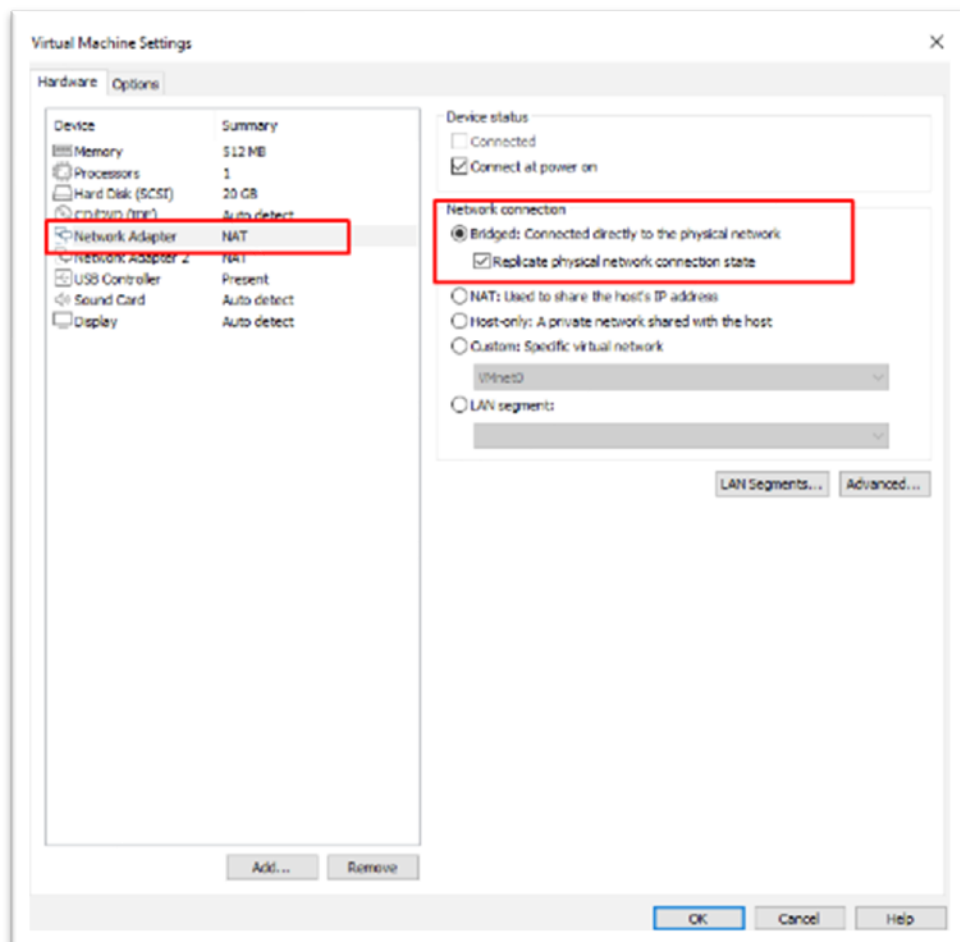
Ajouter une deuxième carte réseau
Cliquer sur le bouton "Add..."



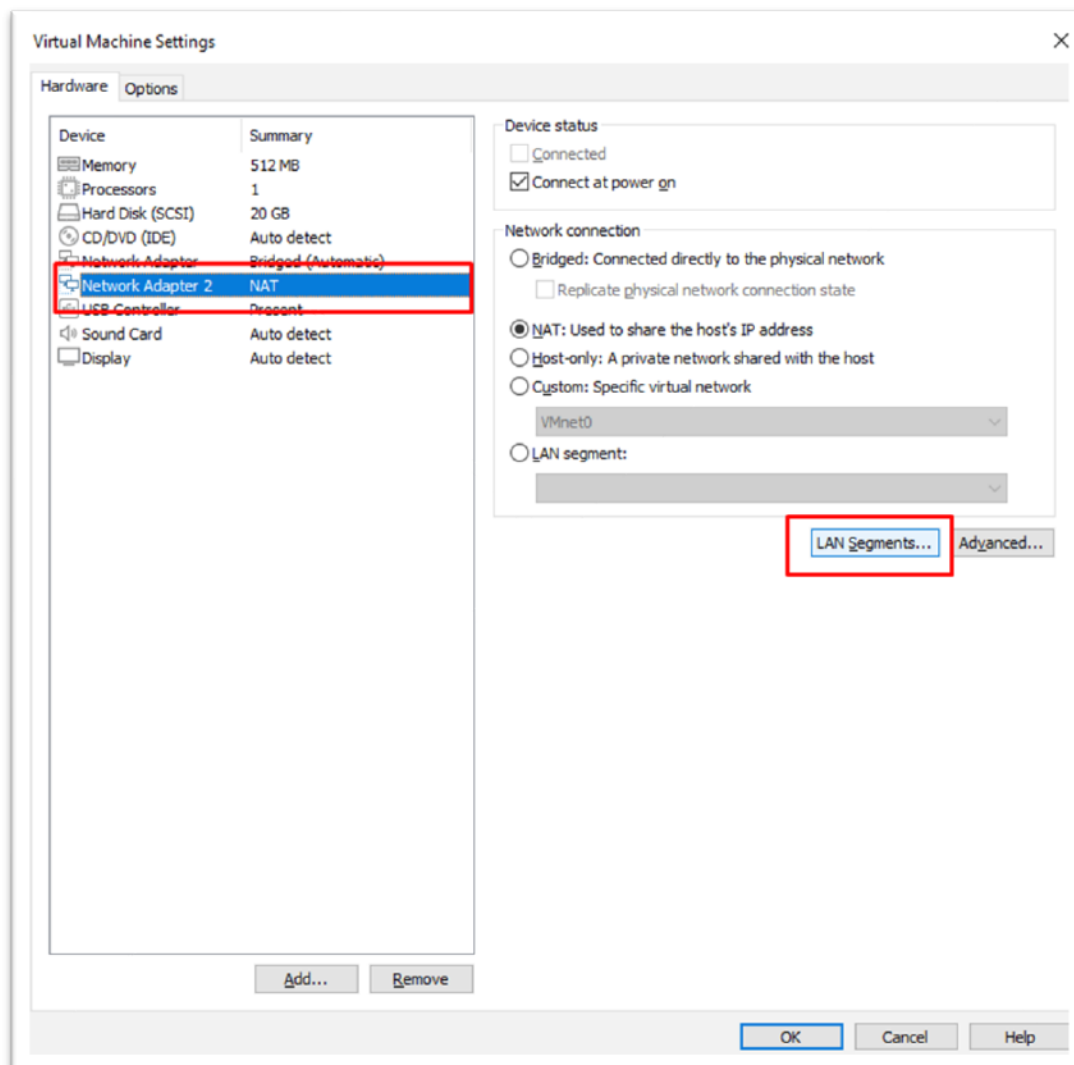
Sélectionner Network Adapter
Cliquer sur "Finish"



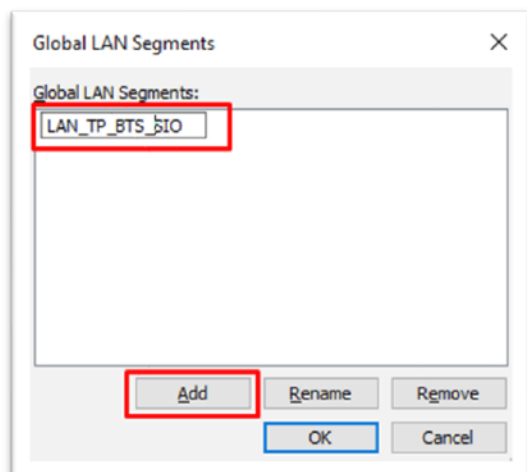
Sur la première carte réseau, "Network Adapter", Définir la rubrique "Network connection" sur "Bridged" et cochez l'option "Replicate physical network connection state".



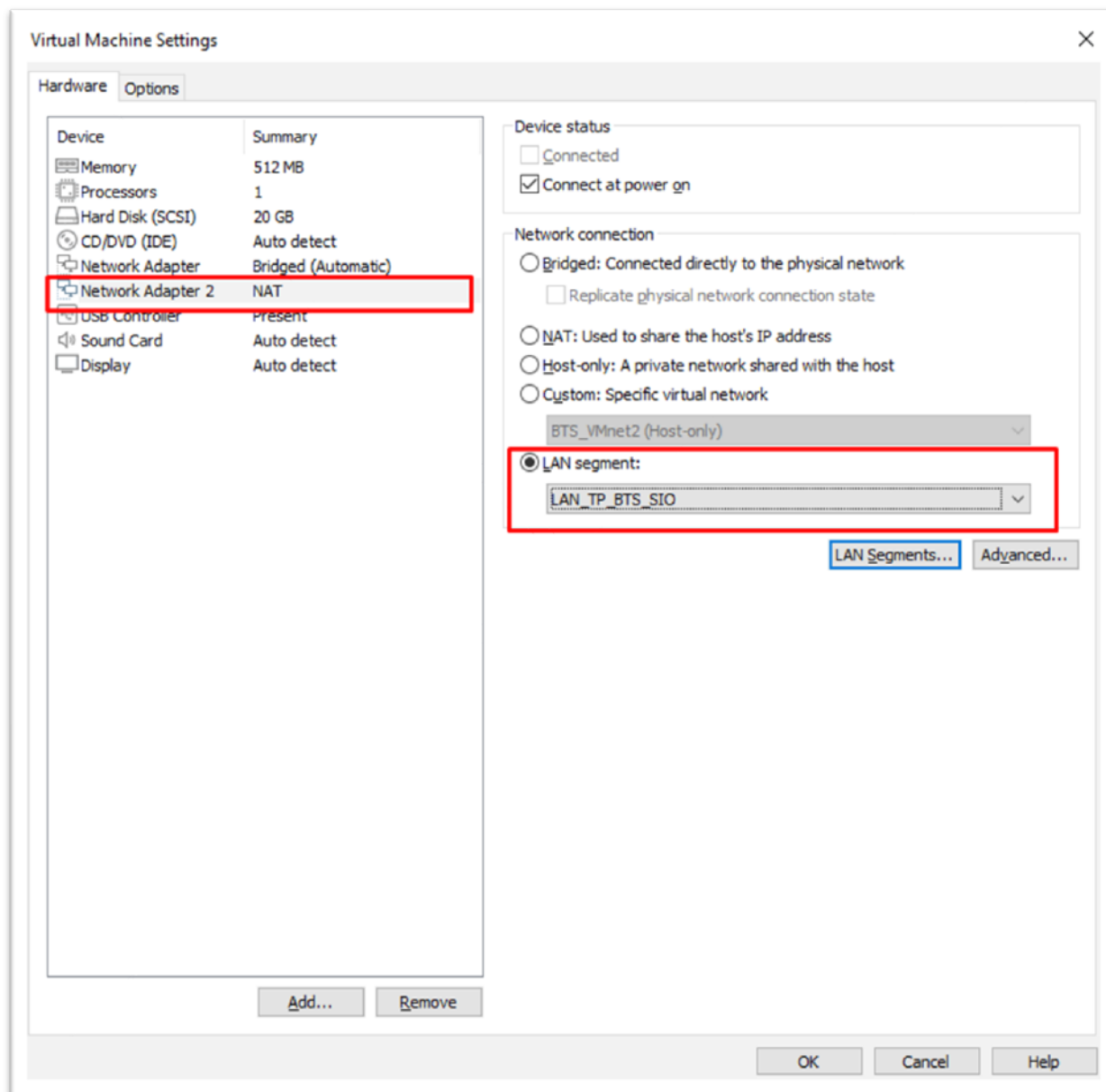
Sélectionner la deuxième carte réseau, "Network Adapter 2", et cliquer sur le bouton "Lan Segment..."



Cliquer sur "Add" et nommer le segment réseau. Ici, "LAN TP BTS SIO"



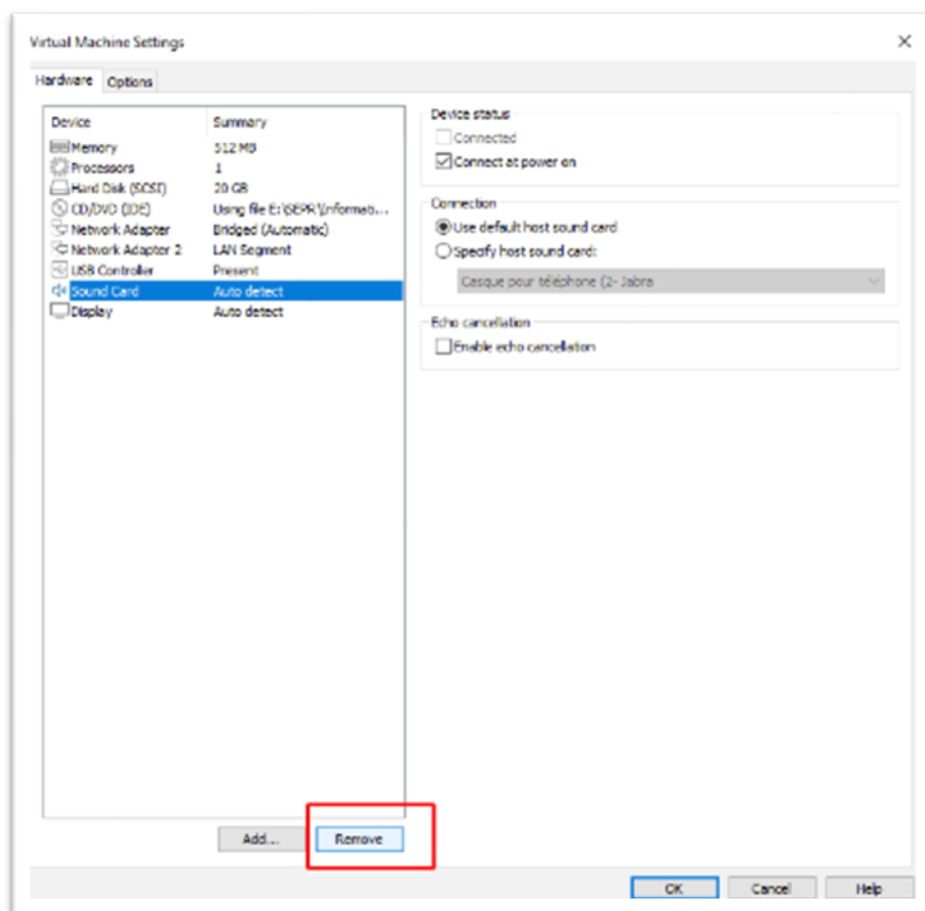
Attribuer ensuite au "Network Adapter 2" l'option "LAN segment" et sélectionner le segment créé dans l'étape précédente. Ici "LAN_TP_BTS_SIO"



Remarque importante :

Il faudra bien connecter les cartes réseaux de toutes les autres machines virtuelles sur ce même segment LAN.

Supprimer la carte son en sélectionnant "Sound Card" puis en cliquant sur le bouton "Remove"

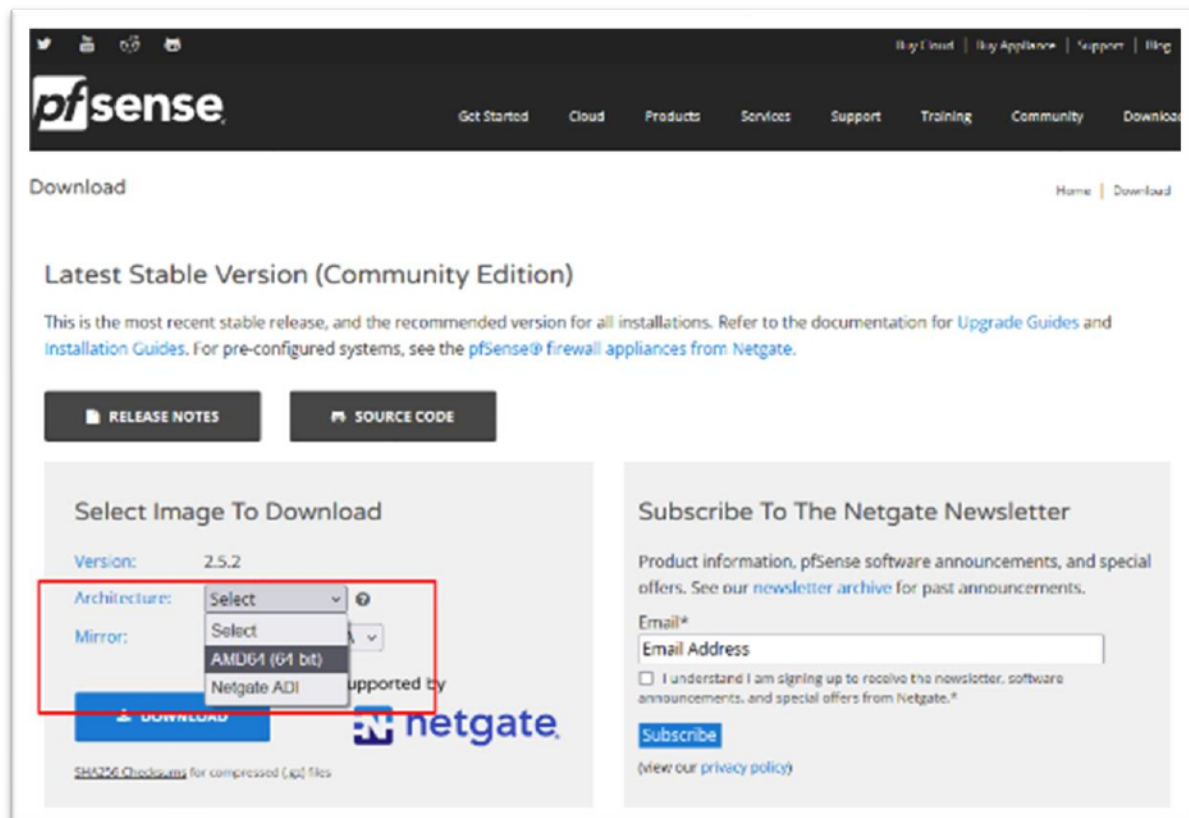


Installation de PfSense

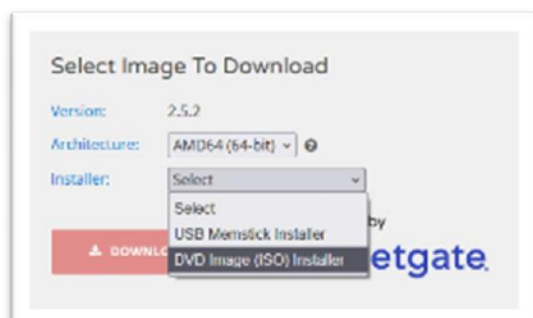
Téléchargement et installation du programme sur la machine virtuelle

Télécharger PfSense sur la page <https://www.pfsense.org/download/>

Sélectionner l'architecture "AMD (64-bit)"



Sélectionner l'installateur "DVD Image (ISO)"



Sélectionner le serveur "Frankfurt, Germany"



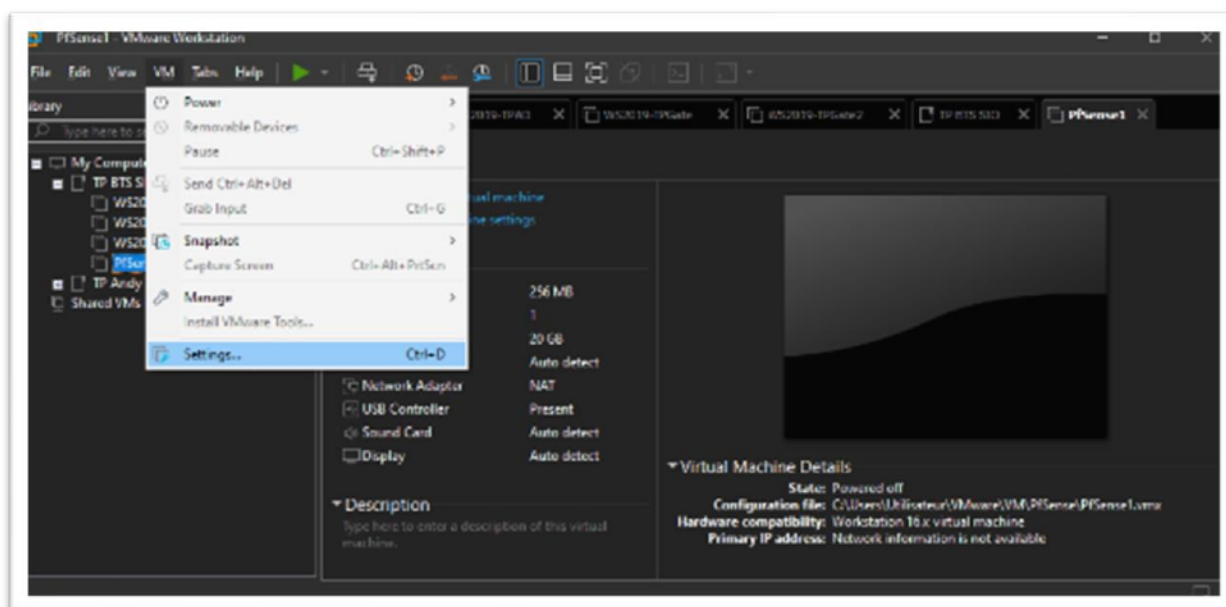
Et cliquer sur le bouton "Download" qui s'est coloré en bleu.

Bien penser à décompresser le fichier téléchargé

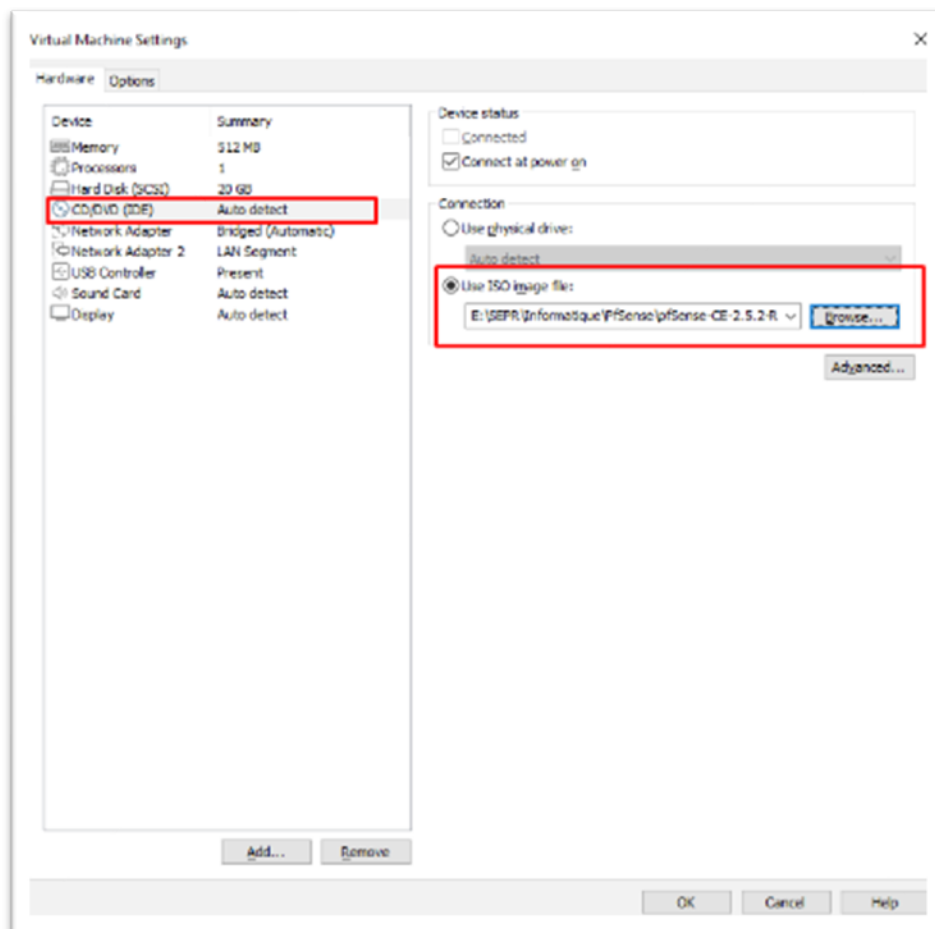
De nouveau, sélectionner la machine virtuelle

Menu "VM"

Sélectionner "Settings..."

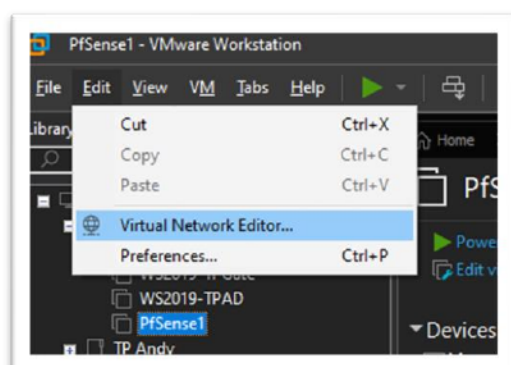


Dans l'onglet "Hardware", sélectionner "CD/DVD", choisir l'option "Use image file" et indiquer le chemin vers le fichier iso.

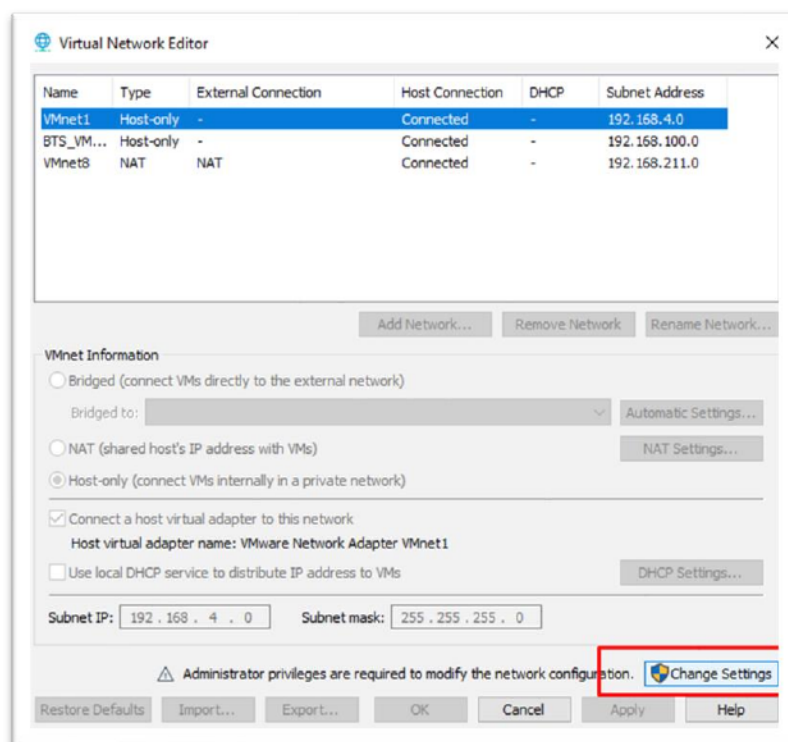


Avant de démarrer la machine virtuelle, vérifier que la carte "Network Adapter" branchée sur la bonne carte réseau de l'ordinateur hôte.

Menu "Edit", sélectionner "Virtual Network Editor..."

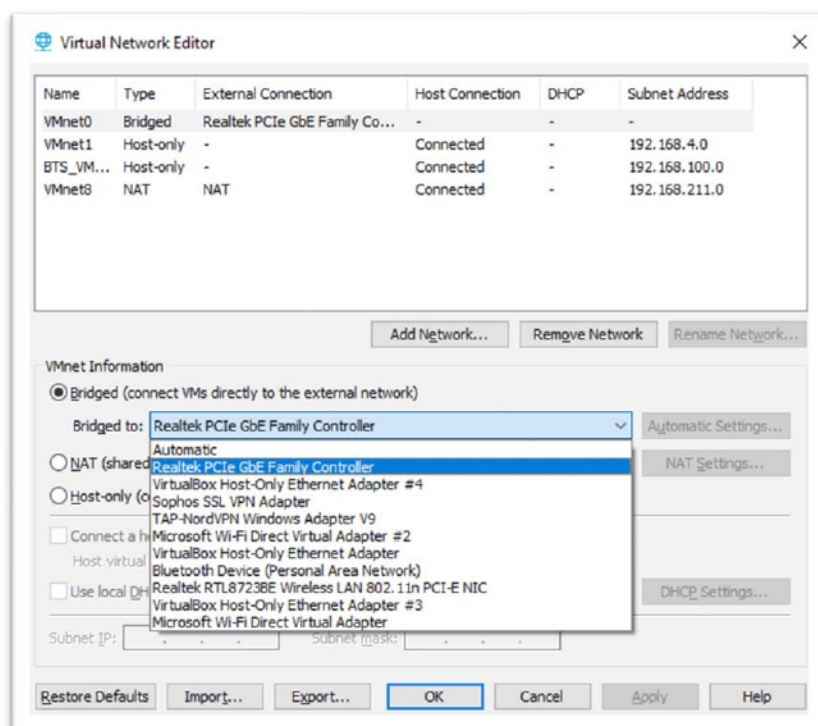


Cliquer sur "Change Settings" et autoriser le programme à prendre les privilèges administrateur.

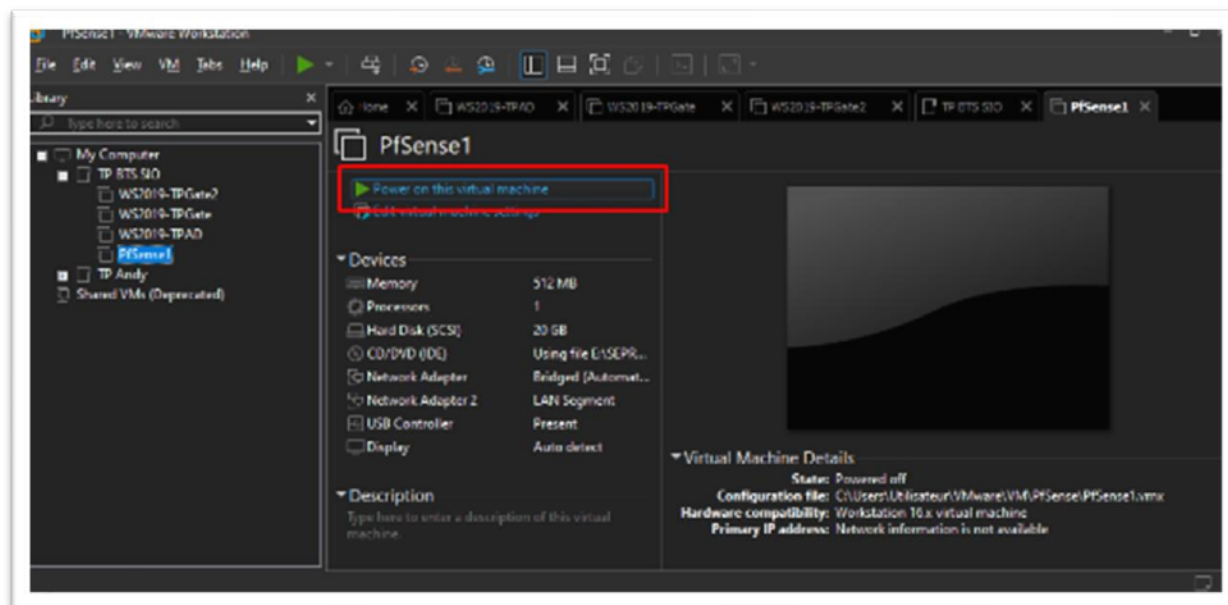


Définir le réseau bridge sur la carte physique connecté à Internet. Ici, je suis connecté en Ethernet donc je choisis ma carte "Realtek PCIe GbE Family Controller".

Si la connexion passe sur un réseau WiFi, il faut penser à mettre à jour cette option en sélectionnant la carte WiFi de l'ordinateur hôte



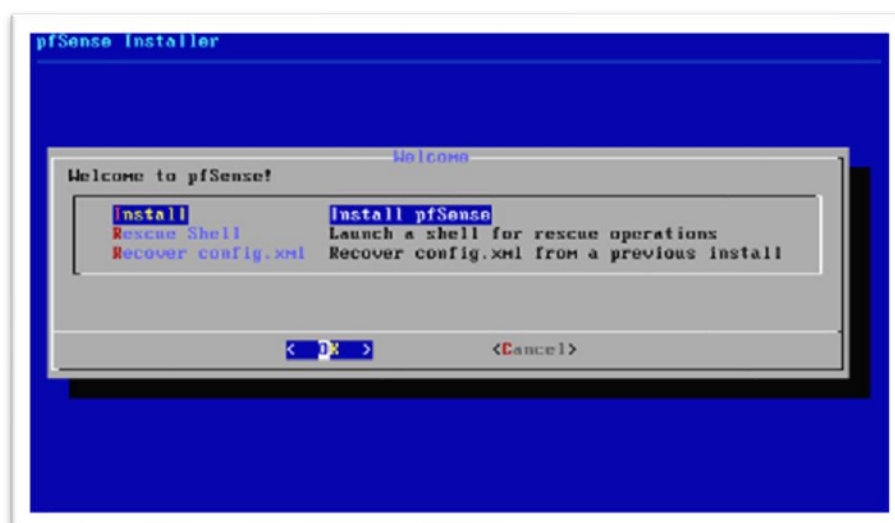
Sélectionner la machine virtuelle et démarrer en cliquant sur "Power on this virtual machine"



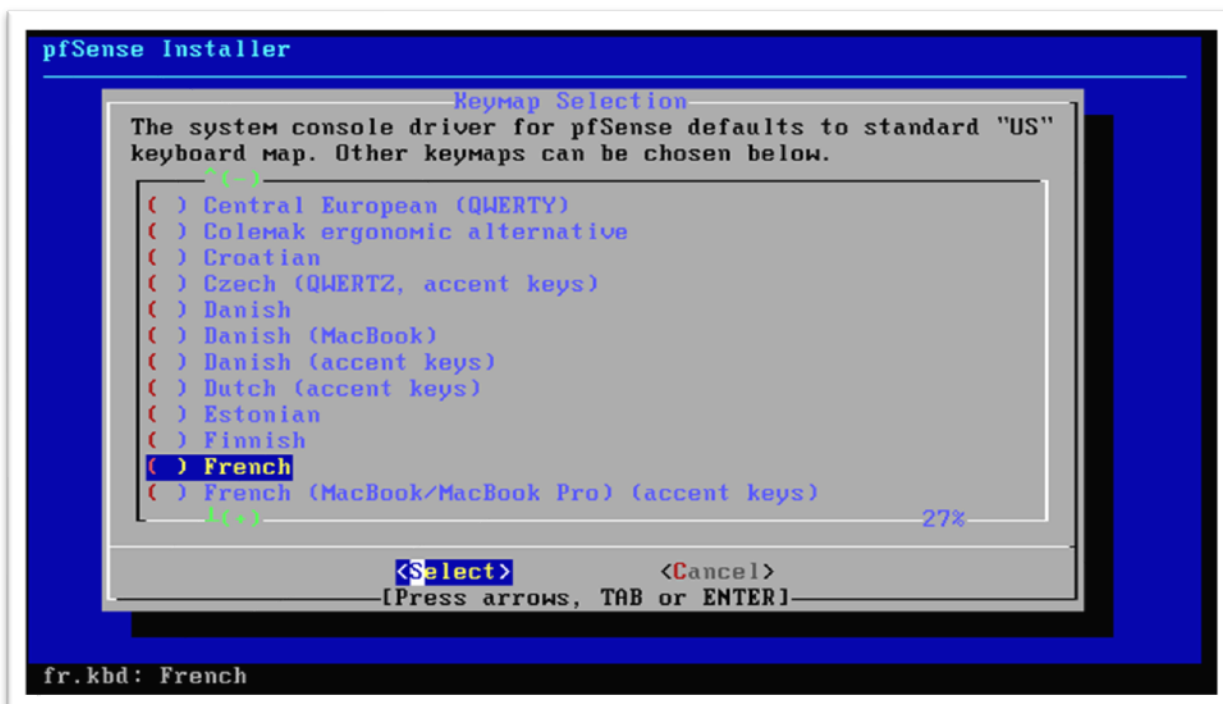
Accepter les conditions en envoyant Entrée sur le mot "Accept"



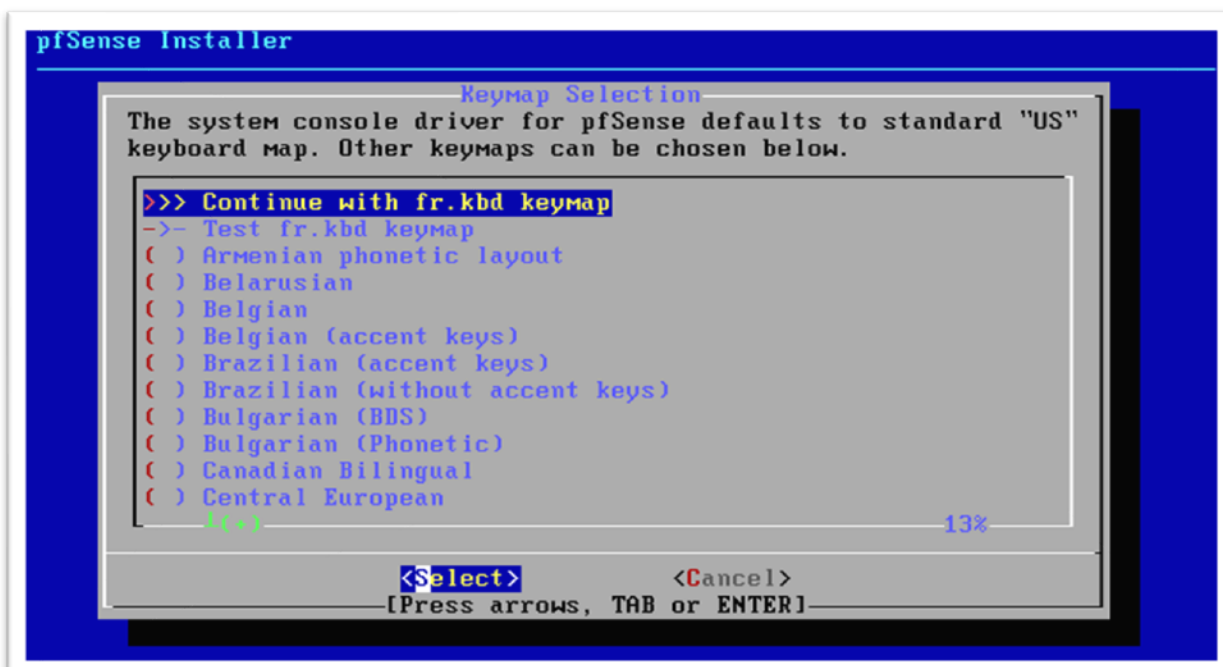
Choisir "Install" et valider



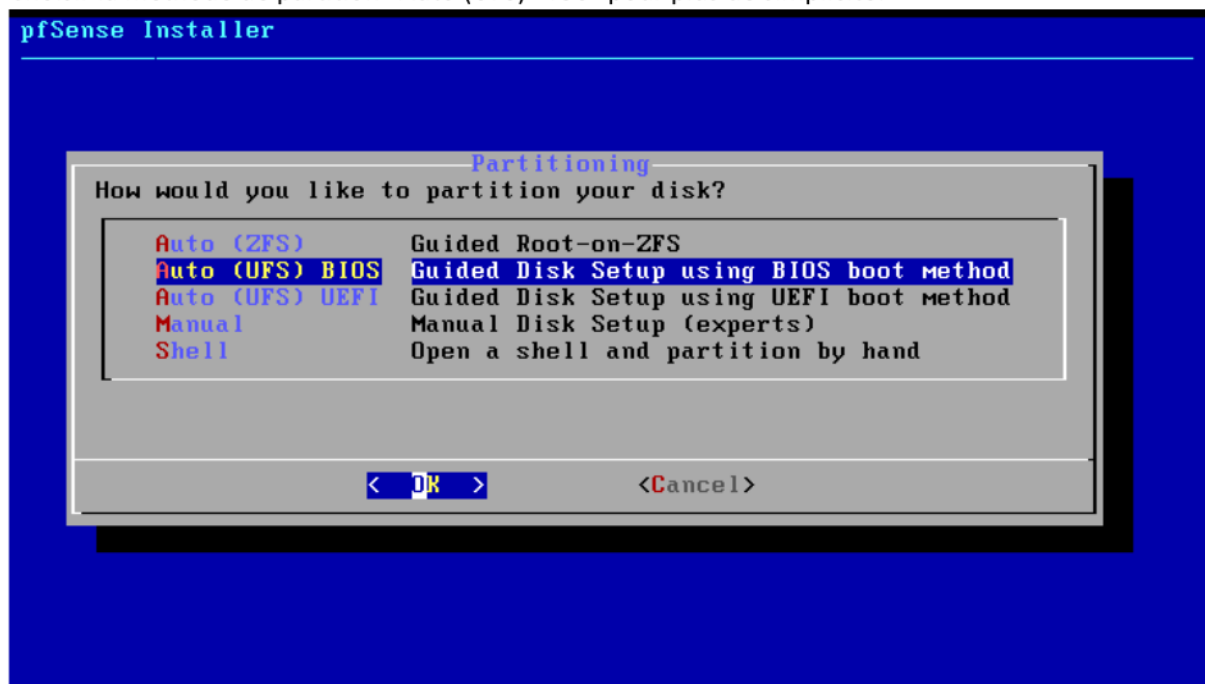
Menu "Keymap Selection", sélectionner la version française du clavier : "French".
Attention, la version French (accent Keys) ne correspond pas au clavier français.



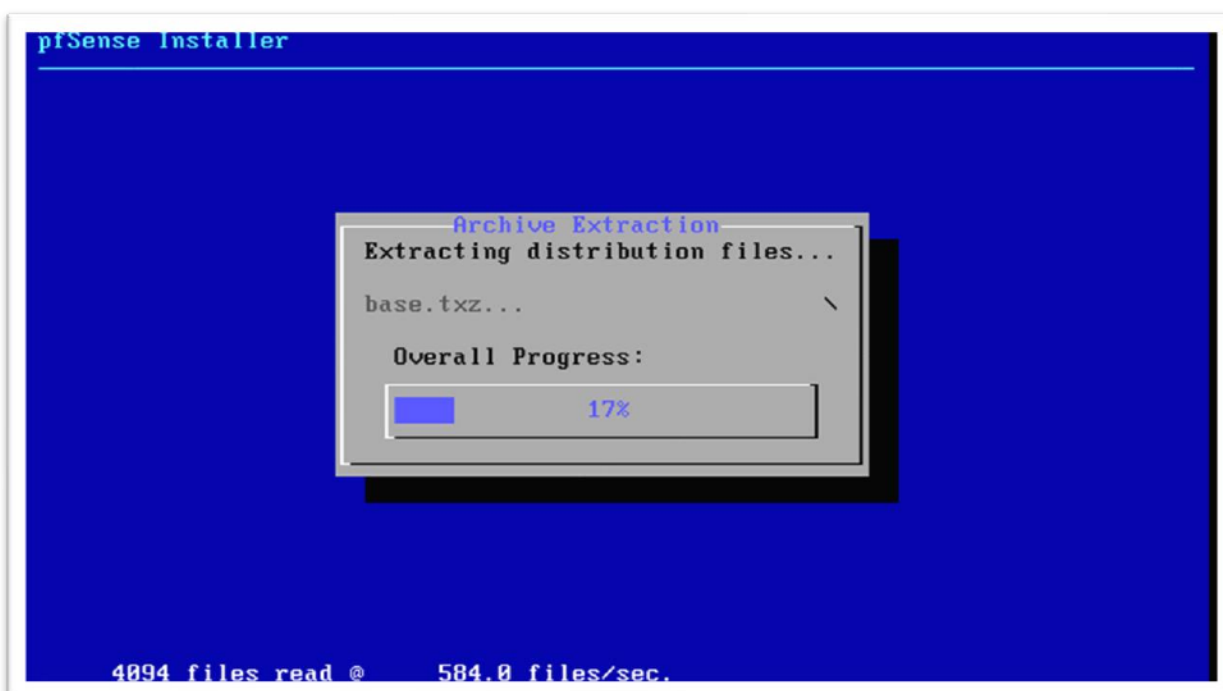
Valider une deuxième fois en sélectionnant "Continue with fr.Kbd keymap"



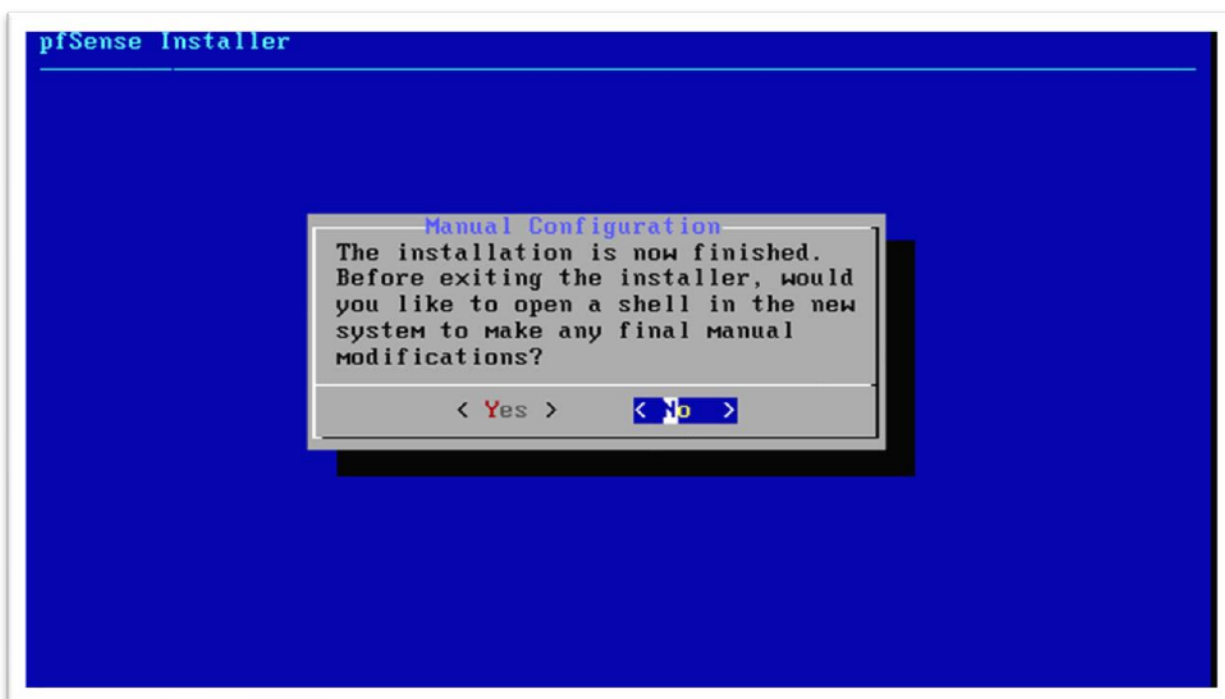
Choisir la méthode de partition "Auto (UFS) BIOS" pour plus de simplicité.



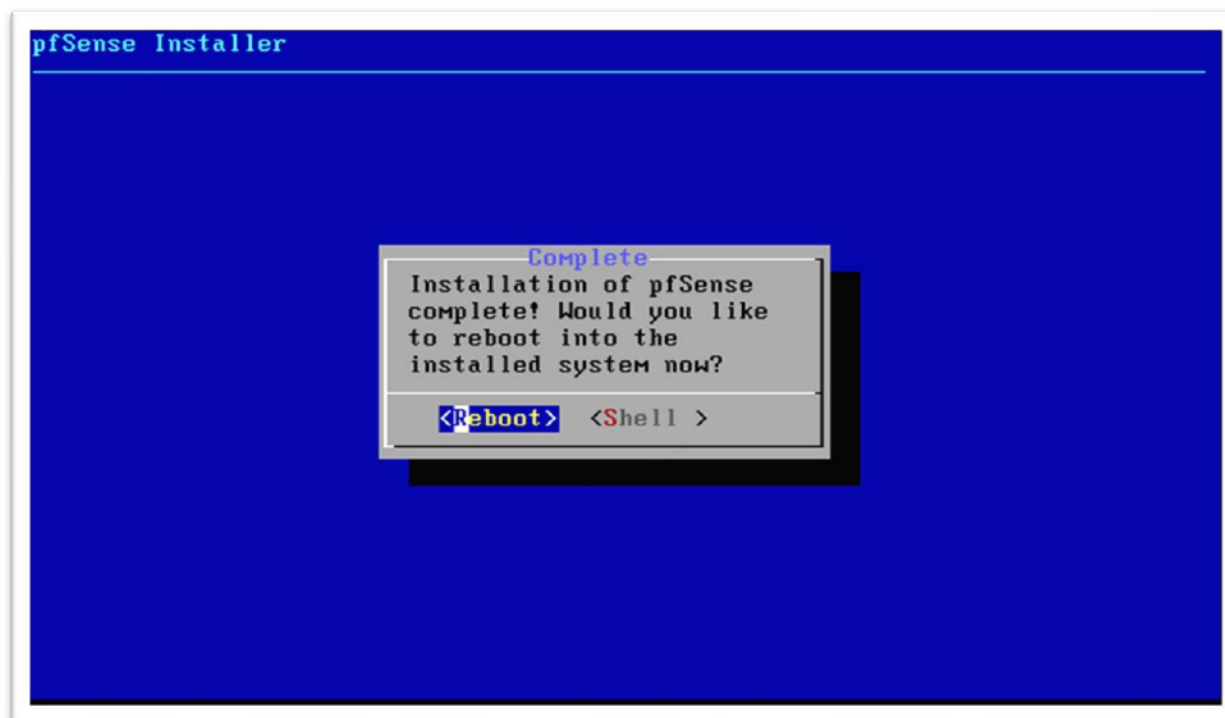
Le programme va s'installer.



L'installation se termine par un message qui demande s'il est besoin d'ouvrir une invite de commande. Choisir "Non".



Redémarrer la machine virtuelle. Sélectionner "Reboot".



Configuration des cartes réseau.

En redémarrant, le système atteint un écran "menu".

Ouvrir un Shell en sélectionnant l'option numéro 8.

```
FreeBSD/amd64 (pfSense.home.arp) (ttyv0)
VMware Virtual Machine - Netgate Device ID: acbd319f0aed1e1a78c0
*** Welcome to pfSense 2.5.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.40/24
                  v6/DHCP6: 2a01:e34:ec18:49c0:20c:29ff:fe39:155
/64
LAN (lan)      -> em1      -> v4: 192.168.100.1/24
                  v6: FD23:11:1980:192:168:100:1:0/64

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: 8

[2.5.2-RELEASE][root@pfSense.home.arp]/root: █
```

Passer le clavier en azerty via la commande suivante :

`kbdcontrol -l /usr/share/syscons/keymaps/fr.iso.kbd`

```
VMware Virtual Machine - Netgate Device ID: acbd319f0aed1e1a78c0
*** Welcome to pfSense 2.5.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.40/24
                  v6/DHCP6: 2a01:e34:ec18:49c0:20c:29ff:fe39:155
/64
LAN (lan)      -> em1      -> v4: 192.168.100.1/24
                  v6: FD23:11:1980:192:168:100:1:0/64

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell

Enter an option: 8

[2.5.2-RELEASE][root@pfSense.home.arp]/root: kbdcontrol -l /usr/share/syscons/k
eymaps/fr.iso.kbd
[2.5.2-RELEASE][root@pfSense.home.arp]/root: █
```

Sortir de l'invite de commande et revenir au menu en tapant "exit"

```

VMware Virtual Machine - Netgate Device ID: acbd319f0aed1e1a78c0
*** Welcome to pfSense 2.5.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.40/24
                                   v6/DHCP6: 2a01:e34:ec18:49c0:20c:29ff:fe39:155
/64
LAN (lan)      -> em1      -> v4: 192.168.100.1/24
                                   v6: FD23:11:1980:192:168:100:1:0/64

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: 8

[2.5.2-RELEASE][root@pfSense.home.arpal]/root: kbdcontrol -l /usr/share/syscons/k
eymaps/fr.iso.kbd
[2.5.2-RELEASE][root@pfSense.home.arpal]/root: exit

```

Choisir l'option 2 "Set interface(s) IP address"

```

pfSense 2.5.2-RELEASE amd64 Fri Jul 02 15:33:00 EDT 2021
Bootup complete

FreeBSD/amd64 (pfSense.home.arpal) (ttyv0)

VMware Virtual Machine - Netgate Device ID: acbd319f0aed1e1a78c0
*** Welcome to pfSense 2.5.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.40/24
                                   v6/DHCP6: 2a01:e34:ec18:49c0:20c:29ff:fe39:155
/64
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: 2

```

Choisir l'interface LAN. Ici, le choix numéro 2

```
*** Welcome to pfSense 2.5.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.40/24
                  v6/DHCP6: 2a01:e34:ec18:49c0:20c:29ff:fe39:155
/64
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults 13) Update from console
5) Reboot system             14) Enable Secure Shell (sshd)
6) Halt system               15) Restore recent configuration
7) Ping host                 16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2
```

Entrer l'adresse IP que l'on veut donner à ce routeur. Ici 192.168.100.1

```
                  v6/DHCP6: 2a01:e34:ec18:49c0:20c:29ff:fe39:155
/64
LAN (lan)      -> em1      -> v4: 192.168.1.1/24

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults 13) Update from console
5) Reboot system             14) Enable Secure Shell (sshd)
6) Halt system               15) Restore recent configuration
7) Ping host                 16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.100.1
```

Entrer l'annotation CIDR du masque de sous-réseau. Ici 24.

```
4) Reset to factory defaults      13) Update from console
5) Reboot system                  14) Enable Secure Shell (sshd)
6) Halt system                    15) Restore recent configuration
7) Ping host                      16) Restart PHP-FPM
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.100.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 31):
> 24
```

Le programme propose d'entrer la passerelle si cette carte réseau point vers un WAN. Ne rien entrer pour passer ce choix car dans le LAN, c'est précisément le PfSense qui servira de passerelle.

```
8) Shell

Enter an option: 2

Available interfaces:

1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.100.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 31):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>
```

Entrer l'adresse IPv6. Utile pour certaines fonctionnalités de Windows Serveur 2019.

Ici : FD23:11:1980:192:168:100:1:0

```
1 - WAN (em0 - dhcp, dhcp6)
2 - LAN (em1 - static)

Enter the number of the interface you wish to configure: 2

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 2

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.100.1

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 31):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Enter the new LAN IPv6 address. Press <ENTER> for none:
> fd23:11:1980:192:168:100:1:0
```

Masque de sous réseau IPv6. Ici : 64

```
Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 31):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Enter the new LAN IPv6 address. Press <ENTER> for none:
> fd23:11:1980:192:168:100:1:0

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. ffff:ffff:ffff:ffff:ffff:ffff:ffff:ff00 = 120
     ffff:ffff:ffff:ffff:ffff:ffff:ffff:0   = 112
     ffff:ffff:ffff:ffff:ffff:ffff:0:0      = 96
     ffff:ffff:ffff:ffff:ffff:0:0:0         = 80
     ffff:ffff:ffff:ffff:0:0:0:0            = 64

Enter the new LAN IPv6 subnet bit count (1 to 127):
> 64
```

De nouveau la question sur la passerelle. Ne rien rentrer.

```

255.0.0.0      = 8

Enter the new LAN IPv4 subnet bit count (1 to 31):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Enter the new LAN IPv6 address. Press <ENTER> for none:
> fd23:11:1980:192:168:100:1:0

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. ffff:ffff:ffff:ffff:ffff:ffff:ffff:ff00 = 120
     ffff:ffff:ffff:ffff:ffff:ffff:ffff:0   = 112
     ffff:ffff:ffff:ffff:ffff:ffff:0:0      = 96
     ffff:ffff:ffff:ffff:ffff:0:0:0         = 80
     ffff:ffff:ffff:ffff:0:0:0:0            = 64

Enter the new LAN IPv6 subnet bit count (1 to 127):
> 64

For a WAN, enter the new LAN IPv6 upstream gateway address.
For a LAN, press <ENTER> for none:
> █

```

Ne pas activer le DHCP sur le LAN. Répondre "n" pour refuser. Idem pour l'IPv6.

```

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Enter the new LAN IPv6 address. Press <ENTER> for none:
> fd23:11:1980:192:168:100:1:0

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. ffff:ffff:ffff:ffff:ffff:ffff:ffff:ff00 = 120
     ffff:ffff:ffff:ffff:ffff:ffff:ffff:0   = 112
     ffff:ffff:ffff:ffff:ffff:ffff:0:0      = 96
     ffff:ffff:ffff:ffff:ffff:0:0:0         = 80
     ffff:ffff:ffff:ffff:0:0:0:0            = 64

Enter the new LAN IPv6 subnet bit count (1 to 127):
> 64

For a WAN, enter the new LAN IPv6 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...

Do you want to enable the DHCP6 server on LAN? (y/n) n █

```


Activer le retour à http en tant que protocole de configuration Web. Entrez : "y" pour "oui"

```
Enter the new LAN IPv6 address. Press <ENTER> for none:
> FD23:11:1980:192:168:100:1:0

Subnet masks are entered as bit counts (as in CIDR notation) in pfSense.
e.g. ffff:ffff:ffff:ffff:ffff:ffff:ffff:ff00 = 120
     ffff:ffff:ffff:ffff:ffff:ffff:ffff:0   = 112
     ffff:ffff:ffff:ffff:ffff:ffff:0:0      = 96
     ffff:ffff:ffff:ffff:ffff:0:0:0         = 80
     ffff:ffff:ffff:ffff:0:0:0:0           = 64

Enter the new LAN IPv6 subnet bit count (1 to 127):
> 64

For a WAN, enter the new LAN IPv6 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) N
Disabling IPv4 DHCPD...

Do you want to enable the DHCP6 server on LAN? (y/n) N
Disabling IPv6 DHCPD...

Do you want to revert to HTTP as the webConfigurator protocol? (y/n) █
```

Un récapitulatif s'affiche. Taper Entrée pour continuer

```
For a WAN, enter the new LAN IPv6 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? (y/n) n
Disabling IPv4 DHCPD...

Do you want to enable the DHCP6 server on LAN? (y/n) n
Disabling IPv6 DHCPD...

Please wait while the changes are saved to LAN...
Reloading filter...
Reloading routing configuration...
DHCPD...

The IPv4 LAN address has been set to 192.168.100.1/24

The IPv6 LAN address has been set to fd23:11:1980:192:168:100:1:0/64
You can now access the webConfigurator by opening the following URL in your web
browser:
    http://192.168.100.1/
    http://[fd23:11:1980:192:168:100:1:0]/

Press <ENTER> to continue. █
```

Avant de se connecter à l'interface web, il faut autoriser les connexions à destination de l'interface WAN. Pour cela, ouvrir un Shell en sélectionnant l'option numéro 8.

```
FreeBSD/amd64 (pfSense.home.arp) (ttyv0)

VMware Virtual Machine - Netgate Device ID: acbd319f0aed1e1a78c0

*** Welcome to pfSense 2.5.2-RELEASE (amd64) on pfSense ***

WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.40/24
                                   v6/DHCP6: 2a01:e34:ec18:49c0:20c:29ff:fe39:155
/64
LAN (lan)      -> em1      -> v4: 192.168.100.1/24
                                   v6: FD23:11:1980:192:168:100:1:0/64

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system               14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: 8

[2.5.2-RELEASE][root@pfSense.home.arp]/root: █
```

Autoriser l'accès à l'interface web depuis le WAN via cette commande :
pfSsh.php playback enableallowallwan

```
                                   v6/DHCP6: 2a01:e34:ec18:49c0:20c:29ff:fe39:155
/64
LAN (lan)      -> em1      -> v4: 192.168.100.1/24
                                   v6: FD23:11:1980:192:168:100:1:0/64

0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults    13) Update from console
5) Reboot system               14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
8) Shell

Enter an option: 8

[2.5.2-RELEASE][root@pfSense.home.arp]/root: pfSsh.php playback enableallowallw
an
Adding allow all rule...
Turning off block private networks (if on)...
Turning off block bogon networks (if on)...
Reloading the filter configuration...

[2.5.2-RELEASE][root@pfSense.home.arp]/root: █
```

sortir de l'invite de commande en tapant "exit"

Configuration de base de PfSense

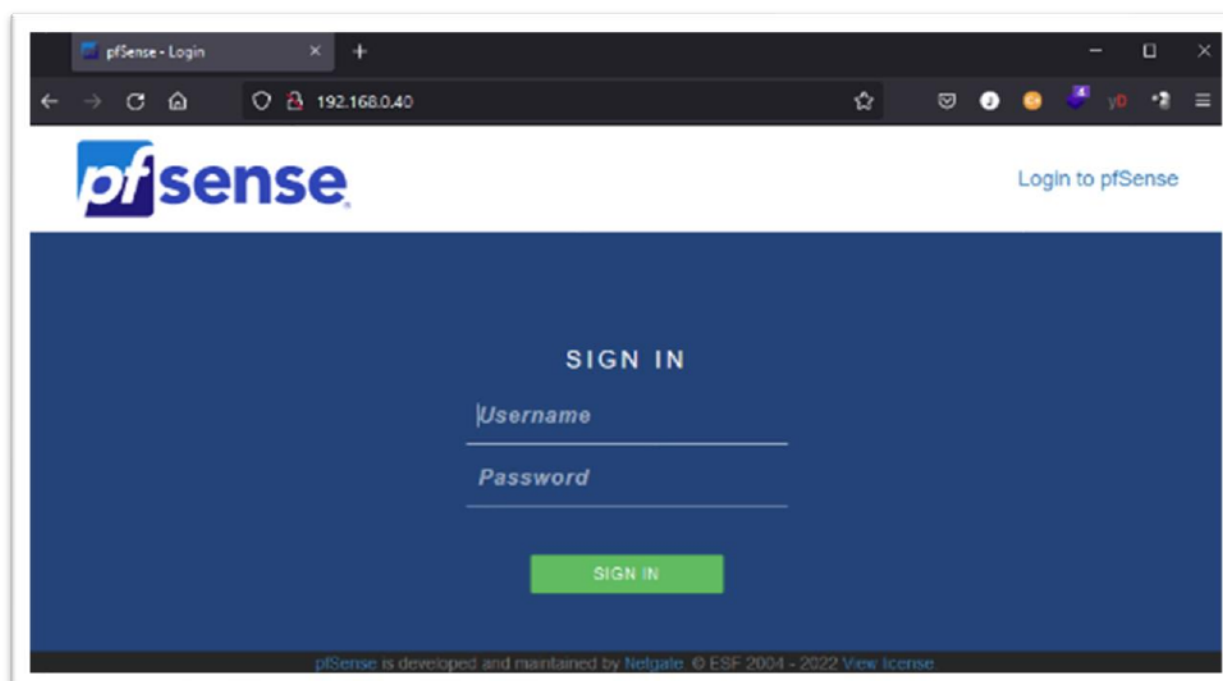
Depuis un navigateur web de l'ordinateur hôte, atteindre l'adresse de l'interface WAN.

L'adresse IP de la machine virtuelle PfSense est affichée au-dessus du menu de celle-ci.

```
*** Welcome to pfSense 2.5.2-RELEASE (amd64) on pfSense ***
WAN (wan)      -> em0      -> v4/DHCP4: 192.168.0.40/24
                                v6/DHCP6: 2a01:e34:ec18:49c8:28c:29ff:fe39:15
64
LAN (lan)      -> em1      -> v4: 192.168.100.1/24
                                v6: FD23:11:1980:192:168:100:1:0/64

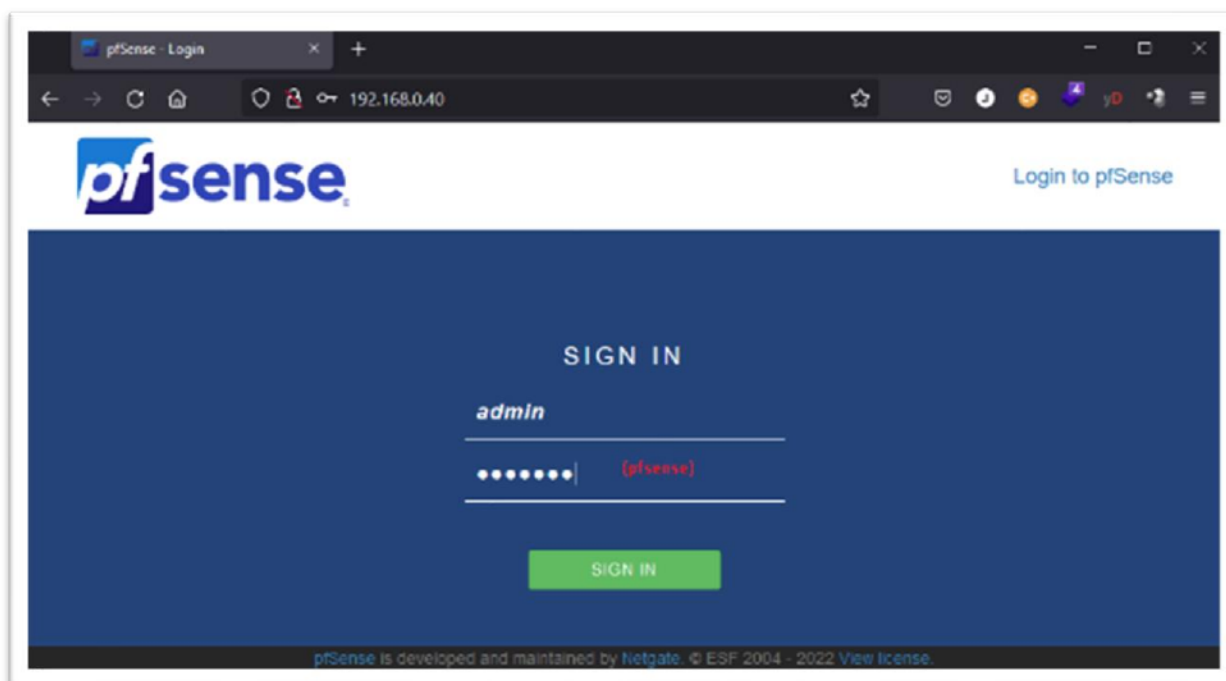
0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                 15) Restore recent configuration
7) Ping host                   16) Restart PHP-FPM
```

Ici 192.168.0.40

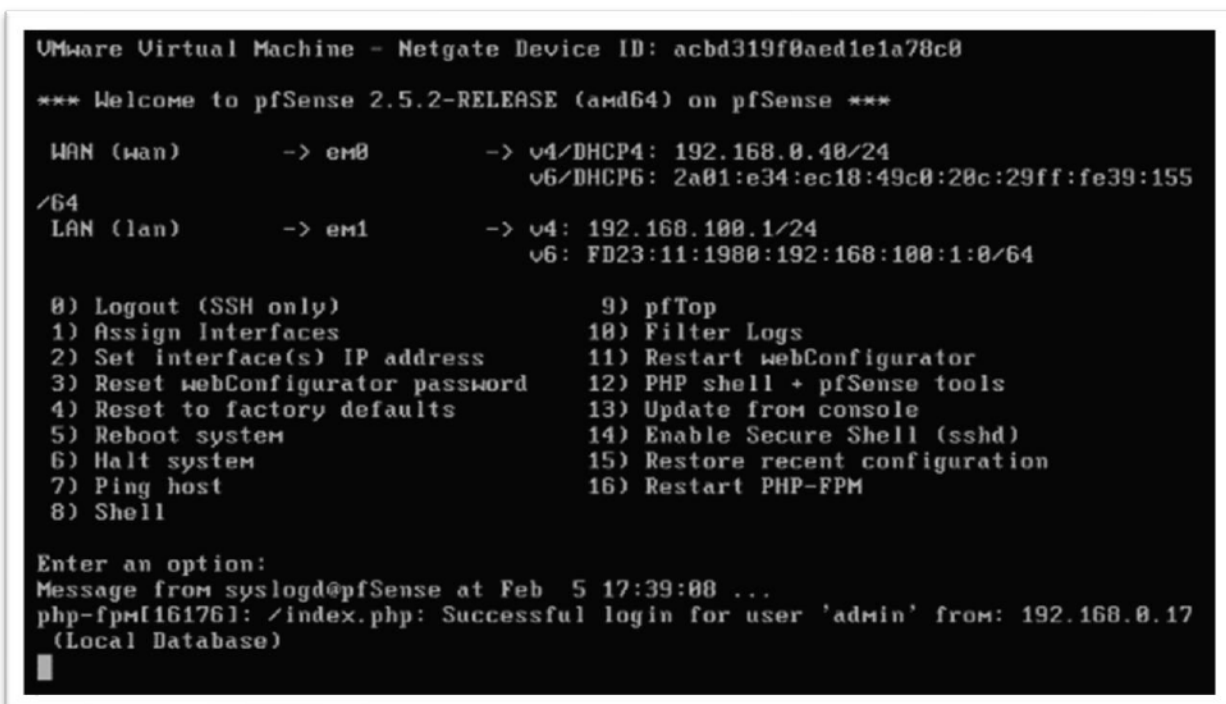


Se connecter avec le compte par défaut :

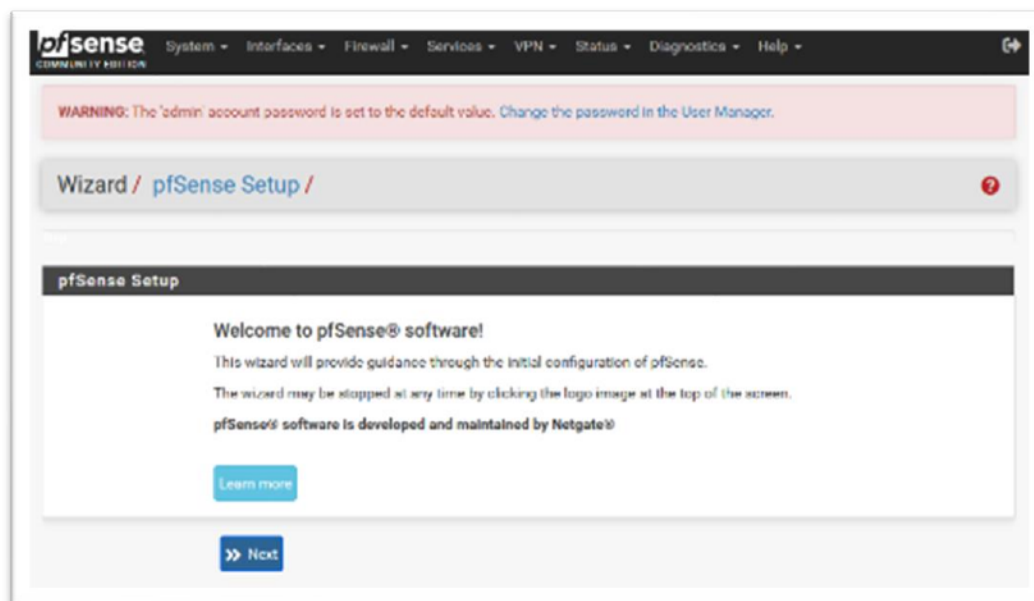
- utilisateur : admin
- mot de passe : pfsense



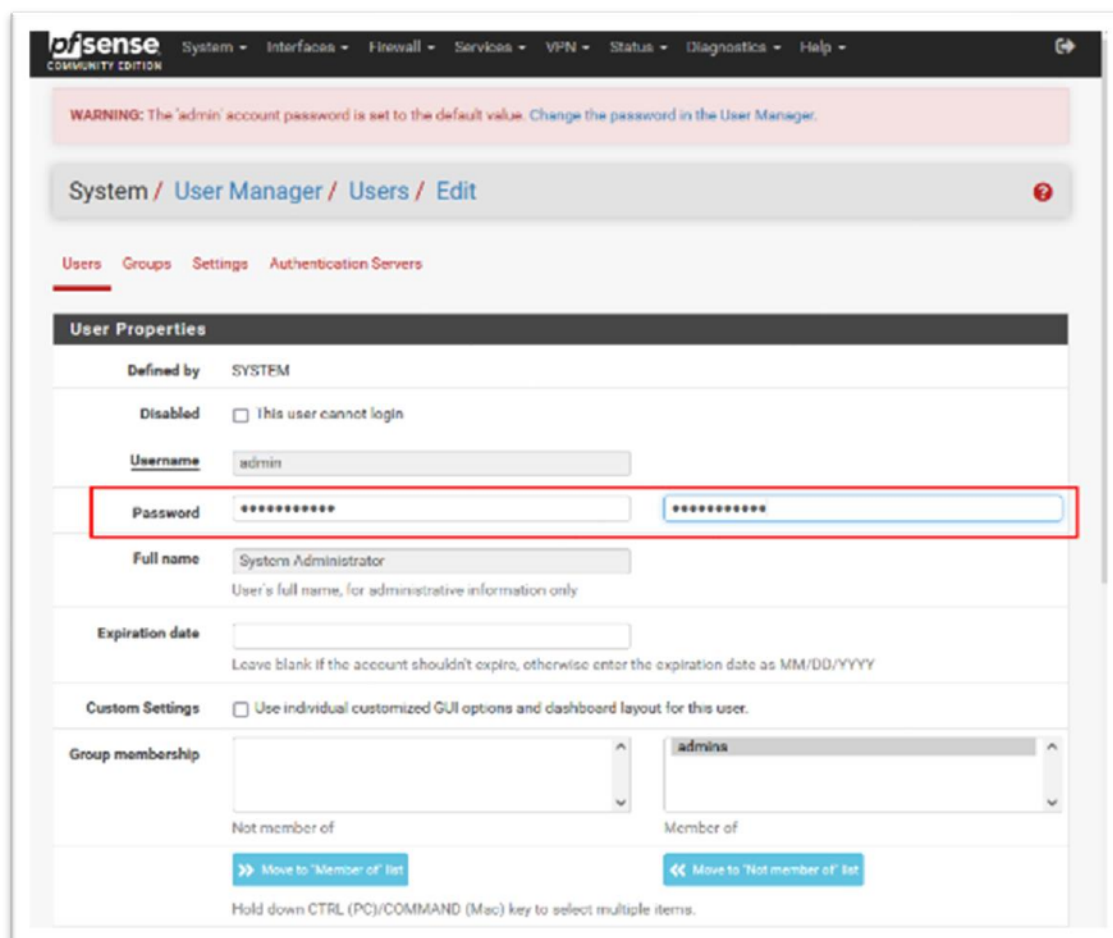
Sur la machine virtuelle, un message indique une connexion réussie



L'interface web envoie la navigation vers la procédure de configuration Wizard Setup. En haut de la page, un message conseille de changer le mot de passe par défaut. Cliquer sur le lien "Change the password in the User Manager."



Entrer un nouveau mot de passe. Ici, WS2019-TPAD



Sauvegarder en cliquant sur le bouton save, en bas de la page

Inherited from	Name	Description	Action
admins	WebCfgr - All pages	Allow access to all pages (admin privilege)	
	User - System: Shell account access	Indicates whether the user is able to login for example via SSH. (admin privilege)	

Security notice: This user effectively has administrator-level access

+ Add

User Certificates

Name	CA

+ Add

Keys

Authorized SSH Keys

Enter authorized SSH keys for this user

IPsec Pre-Shared Key

Save

Cliquer sur le logo PfSense pour atteindre la page d'accueil

System / User Manager / Users

Users Groups Settings Authentication Servers

Username	Full name	Status	Groups	Actions
admin	System Administrator	✓	admins	

+ Add Edit Delete

Le tableau de bord de PfSense inclut une liste d'information sur la configuration matérielle.

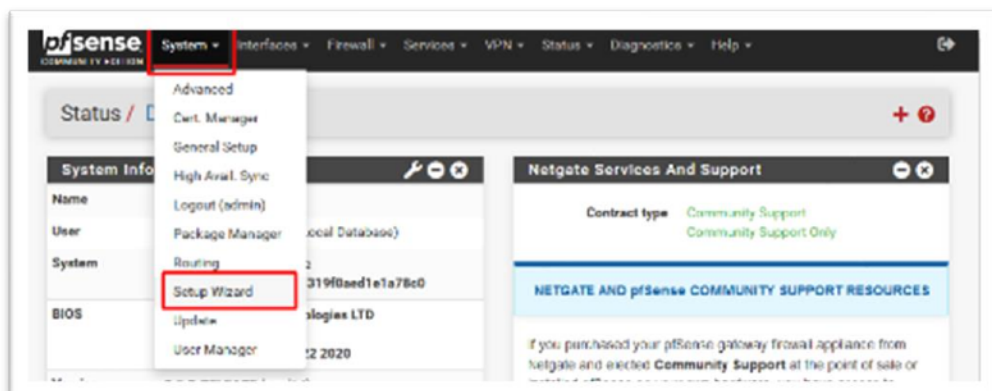
The screenshot displays the pfSense Status Dashboard. The top navigation bar includes links for System, Interfaces, Firewall, Services, VPN, Status, Diagnostics, and Help. The main content area is divided into several sections:

- Status / Dashboard:** The main heading for the current view.
- System Information:** A table providing details about the system:

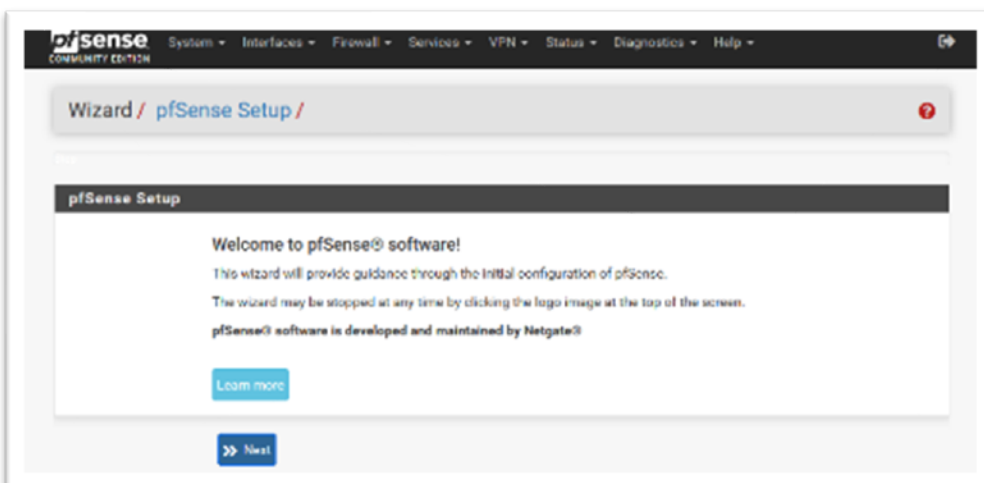
Name	pfSense.home.apa
User	admin@192.168.0.17 (Local Database)
System	VMware Virtual Machine Netgate Device ID: acbd319f0aed1e1a78e0
BIOS	Vendor: Phoenix Technologies LTD Version: 6.00 Release Date: Wed Jul 22 2020
Version	2.5.2-RELEASE (amd64) built on Fri Jul 02 15:33:00 EDT 2021 FreeBSD 12.2-STABLE The system is on the latest version. Version information updated at Sun Feb 6 11:39:25 UTC 2022
CPU Type	Intel(R) Core(TM) i7-6700HQ CPU @ 2.60GHz AES-NI CPU Crypto: Yes (inactive) QAT Crypto: No
Hardware crypto	
Kernel PTI	Enabled
MDS Mitigation	Inactive
Uptime	00 Hour 48 Minutes 04 Seconds
Current date/time	Sun Feb 6 11:47:11 UTC 2022
DNS server(s)	127.0.0.1 192.168.0.254
Last config change	Sun Feb 6 11:40:26 UTC 2022
State table size	0% (8/19000) Show status
MBUF Usage	0% (1414/1000000)
Load average	0.60, 0.53, 0.73
CPU usage	3%
Memory usage	64% of 194 MB
SWAP usage	2% of 1023 MB
Disk usage:	
/	7% of 18GiB - ufs
/var/run	4% of 3.4MiB - ufs in RAM
- Netgate Services And Support:** A section providing information about support resources:
 - Contract type: Community Support (Community Support Only)
 - NETGATE AND pfSense COMMUNITY SUPPORT RESOURCES**
 - Text: "If you purchased your pfSense gateway firewall appliance from Netgate and elected **Community Support** at the point of sale or installed pfSense on your own hardware, you have access to various community support resources. This includes the **NETGATE RESOURCE LIBRARY**."
 - [Upgrade Your Support](#)
 - [Community Support Resources](#)
 - [Netgate Global Support FAQ](#)
 - [Official pfSense Training by Netgate](#)
 - [Netgate Professional Services](#)
 - [Visit Netgate.com](#)
 - Text: "If you decide to purchase a Netgate Global TAC Support subscription, you **MUST** have your **Netgate Device ID (NDI)** from your firewall in order to validate support for this unit. Write down your NDI and store it in a safe place. You can purchase TAC support [here](#)."
- Interfaces:** A table showing the status of network interfaces:

WAN	1000baseT <full-duplex>	192.168.0.40 2a01:e34:ee1b:49e0:20e:29ff:fe39:155
LAN	1000baseT <full-duplex>	192.168.100.1 FD23:11:1980:102:168:100:1:0

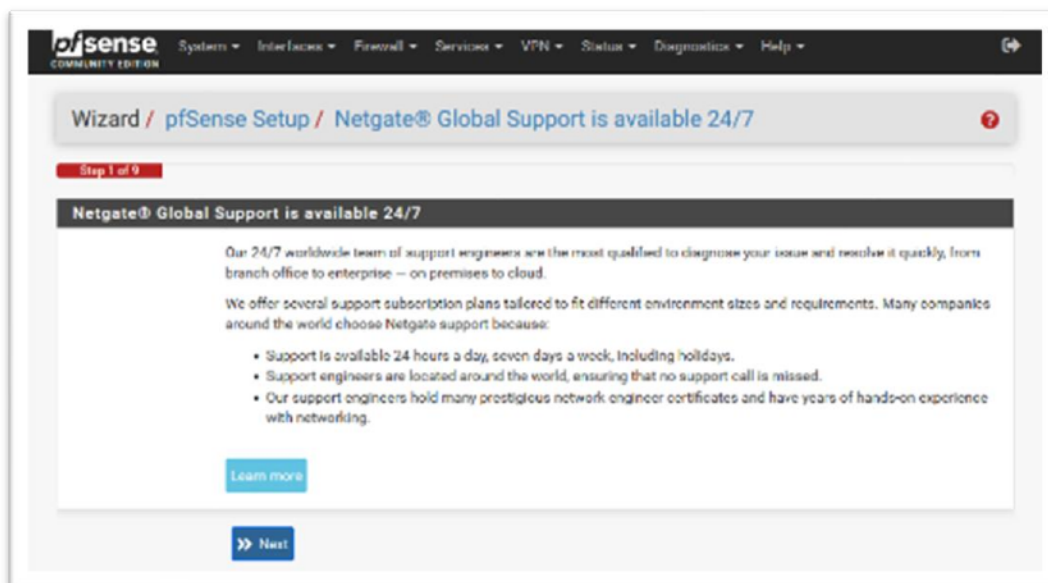
Pour revenir vers la procédure de configuration "Wizard Setup", dérouler le menu "System" et choisir "Setup Wizard".



Cliquer sur "Next"



Puis de nouveau sur "Next"



Renommer la machine virtuelle. Ici, PfSense1
Renseigner le nom du domaine. Ici, bts.local
Renseigner le DNS primaire : Ici, 192.168.100.10
Renseigner le DNS secondaire : Ici, 8.8.8.8
Cliquer sur "Next"

The screenshot shows the 'General Information' step of the pfSense Setup Wizard. The breadcrumb trail is 'Wizard / pfSense Setup / General Information'. A red progress bar indicates 'Step 2 of 9'. The section title is 'General Information'. Below the title, a message states: 'On this screen the general pfSense parameters will be set.' The form contains the following fields:

- Hostname:** A text input field containing 'pfSense1'. Below it, an example is shown: 'EXAMPLE: myserver'.
- Domain:** A text input field containing 'bts.local'. Below it, an example is shown: 'EXAMPLE: mydomain.com'.
- Primary DNS Server:** An empty text input field.
- Secondary DNS Server:** An empty text input field.
- Override DNS:** A checkbox that is checked. Below it, a note reads: 'Allow DNS servers to be overridden by DHCP/PPP on WAN'.

At the bottom of the form is a blue button labeled '>> Next'.

Choisir le fuseau horaire. Ici, Europe/Paris"
Cliquer sur "Next"

The screenshot shows the 'Time Server Information' step of the pfSense Setup Wizard. The breadcrumb trail is 'Wizard / pfSense Setup / Time Server Information'. A red progress bar indicates 'Step 3 of 9'. The section title is 'Time Server Information'. Below the title, a message states: 'Please enter the time, date and time zone.' The form contains the following fields:

- Time server hostname:** A text input field containing '2.pfsense.pool.ntp.org'. Below it, a note reads: 'Enter the hostname (FQDN) of the time server.'
- Timezone:** A dropdown menu with 'Europe/Paris' selected.

At the bottom of the form is a blue button labeled '>> Next'.

Définir une interface WAN "static"

Paramétrer :

Adresse IP : 192.168.0.250

Masque de sous-réseau : 24

Passerelle : 192.168.0.254

Cliquer sur "Next, en bas de la page

pfSense
COMMUNITY EDITION

Wizard / pfSense Setup / Configure WAN Interface

Step 4 of 9

Configure WAN Interface

On this screen the Wide Area Network information will be configured.

SelectedType Static

General configuration

MAC Address

This field can be used to modify ("spoof") the MAC address of the WAN interface (may be required with some cable connections). Enter a MAC address in the following format: xx:xx:xx:xx:xx:xx or leave blank.

MTU

Set the MTU of the WAN interface. If this field is left blank, an MTU of 1492 bytes for PPPoE and 1500 bytes for all other connection types will be assumed.

MSS

If a value is entered in this field, then MSS clamping for TCP connections to the value entered above minus 40 (TCP/IP header size) will be in effect. If this field is left blank, an MSS of 1492 bytes for PPPoE and 1500 bytes for all other connection types will be assumed. This should match the above MTU value in most all cases.

Static IP Configuration

IP Address 192.168.0.250

Subnet Mask 24

Upstream Gateway 192.168.0.254

Paramétrage du réseau LAN

Adresse IP : 192.168.100.1

Masque de sous-réseau : 24

Cliquer sur "Next"

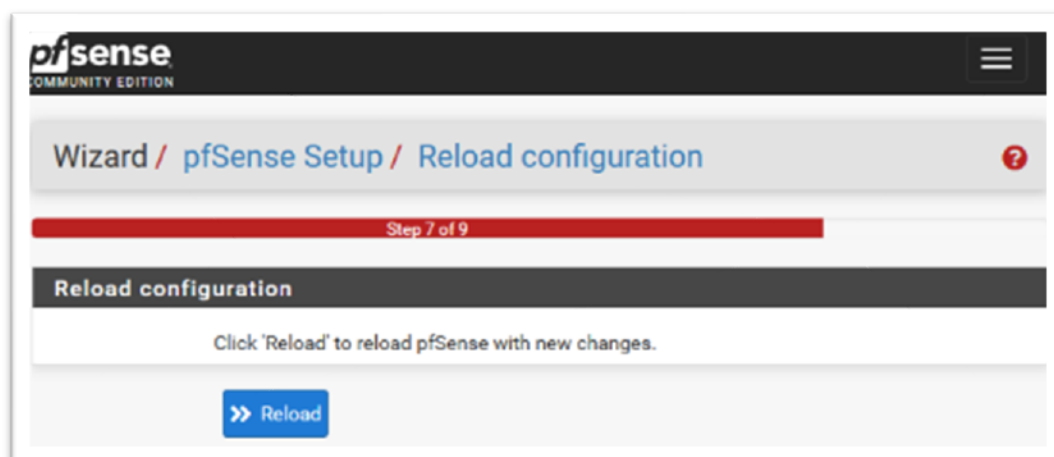
The screenshot shows the 'Configure LAN Interface' step of the pfSense setup wizard. The breadcrumb trail is 'Wizard / pfSense Setup / Configure LAN Interface'. A progress bar indicates 'Step 5 of 9'. The title is 'Configure LAN Interface'. Below the title, a message states: 'On this screen the Local Area Network information will be configured.' There are two input fields: 'LAN IP Address' with the value '192.168.100.1' and a subtext 'Type dhcp if this interface uses DHCP to obtain its IP address.', and 'Subnet Mask' with the value '24'. A blue 'Next' button is at the bottom.

Entrer le mot de passe administrateur

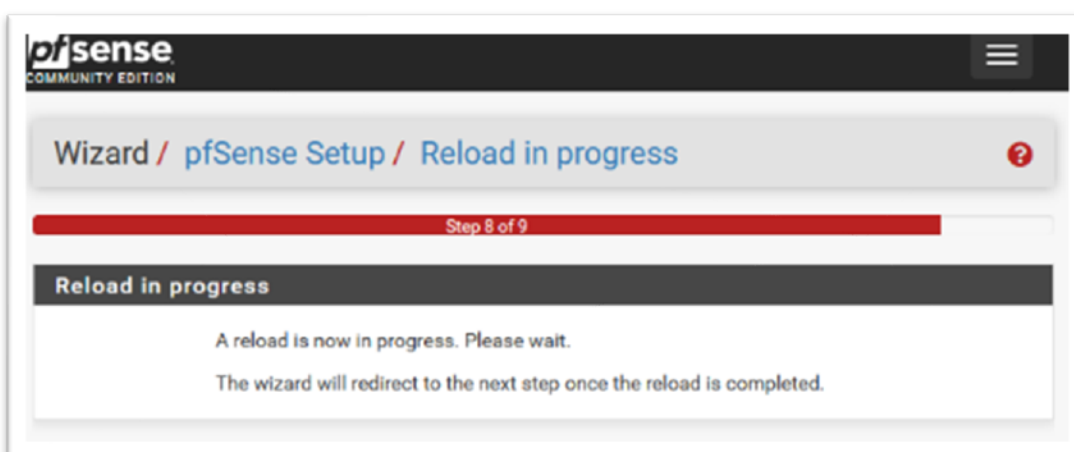
Cliquer sur "Next"

The screenshot shows the 'Set Admin WebGUI Password' step of the pfSense setup wizard. The breadcrumb trail is 'Wizard / pfSense Setup / Set Admin WebGUI Password'. A progress bar indicates 'Step 6 of 9'. The title is 'Set Admin WebGUI Password'. Below the title, a message states: 'On this screen the admin password will be set, which is used to access the WebGUI and also SSH services if enabled.' There are two password input fields: 'Admin Password' (masked with dots) and 'Admin Password AGAIN'. A blue 'Next' button is at the bottom.

Cliquer sur "Reload" pour charger la nouvelle configuration

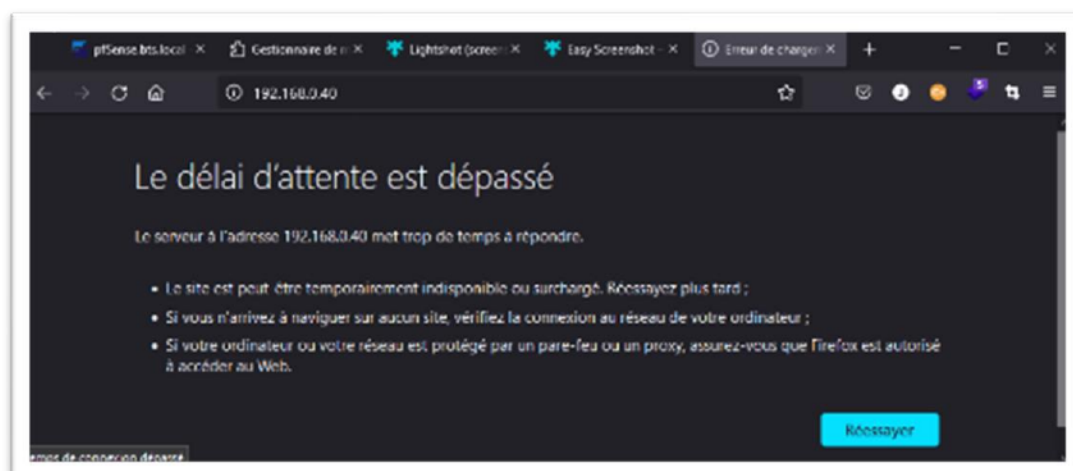


Une page demande à l'utilisateur de patienter



Le chargement de la page échoue car l'adresse IP "WAN" du routeur a changé.

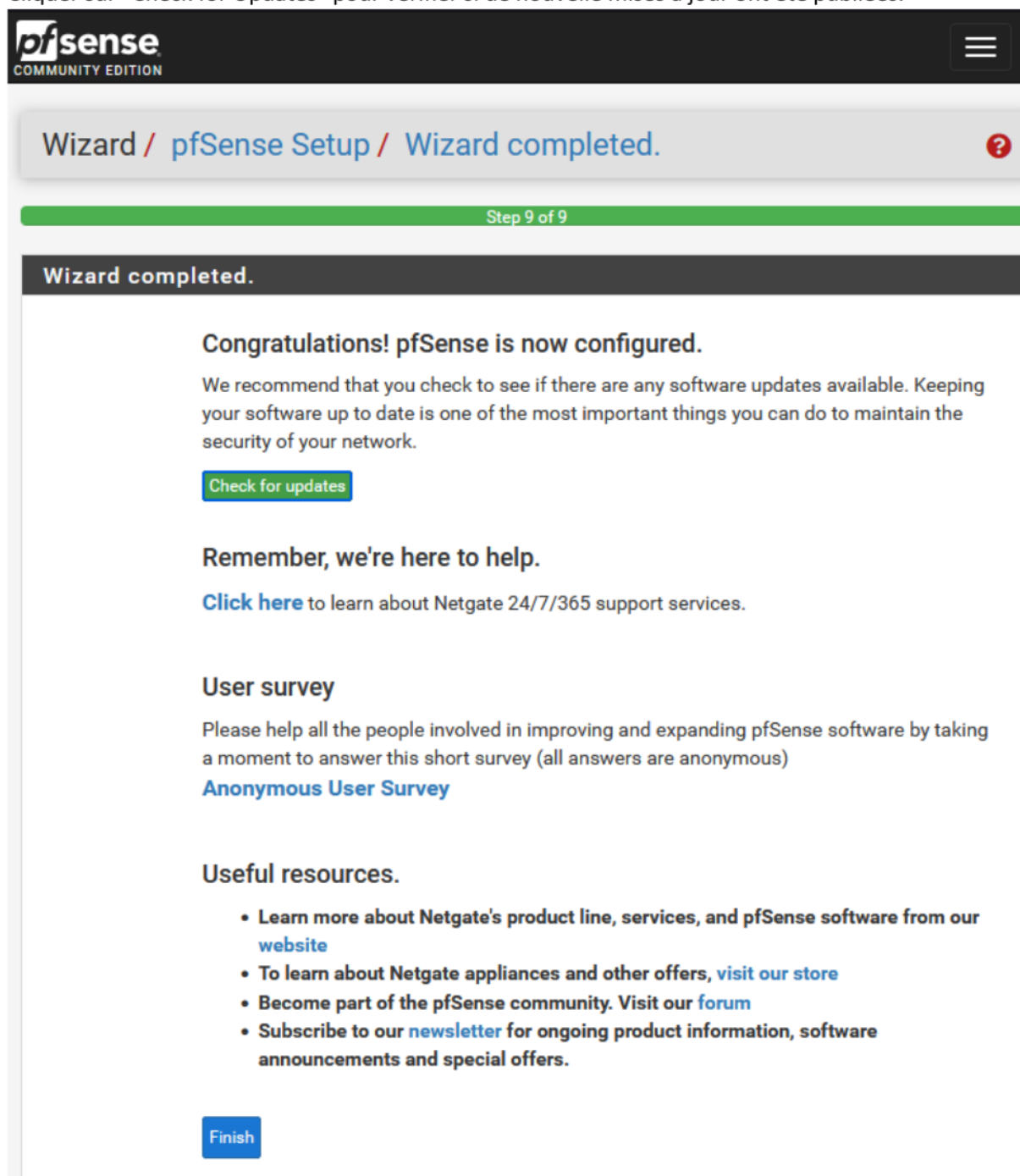
Recharger la page en changeant la nouvelle adresse IP. Ici, 192.168.0.250



Relancer la procédure Wizard Setup.

La configuration aboutit. Une page invite l'utilisateur à vérifier les mises à jour ou à terminer la procédure.

Cliquer sur "Check for Updates" pour vérifier si de nouvelles mises à jour ont été publiées.



The screenshot shows the pfSense Community Edition interface. At the top, the pfSense logo and 'COMMUNITY EDITION' are visible. A breadcrumb trail reads 'Wizard / pfSense Setup / Wizard completed.' with a red question mark icon. Below this, a green progress bar indicates 'Step 9 of 9'. A dark grey header bar contains the text 'Wizard completed.'.

Congratulations! pfSense is now configured.

We recommend that you check to see if there are any software updates available. Keeping your software up to date is one of the most important things you can do to maintain the security of your network.

[Check for updates](#)

Remember, we're here to help.

[Click here](#) to learn about Netgate 24/7/365 support services.

User survey

Please help all the people involved in improving and expanding pfSense software by taking a moment to answer this short survey (all answers are anonymous)

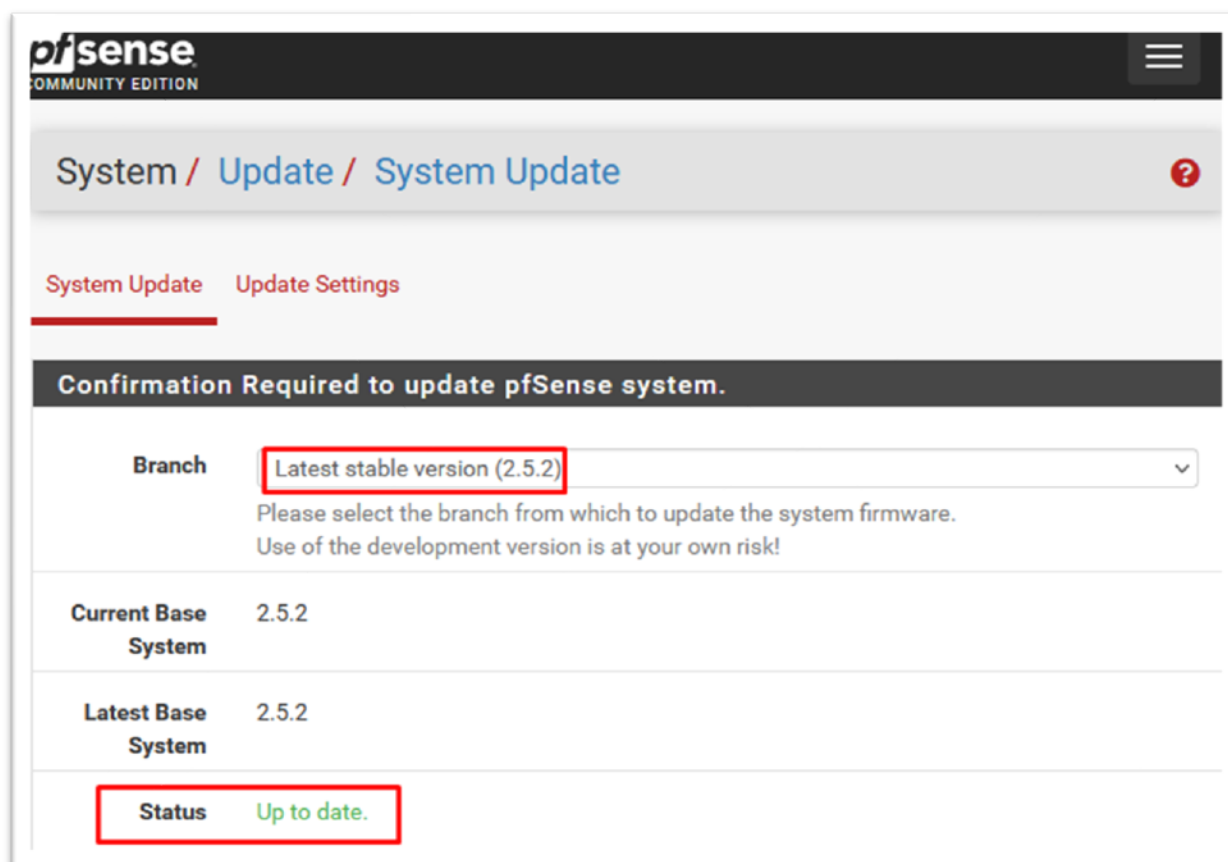
[Anonymous User Survey](#)

Useful resources.

- Learn more about Netgate's product line, services, and pfSense software from our [website](#)
- To learn about Netgate appliances and other offers, [visit our store](#)
- Become part of the pfSense community. Visit our [forum](#)
- Subscribe to our [newsletter](#) for ongoing product information, software announcements and special offers.

[Finish](#)

Ici, la version est bien à jour.



The screenshot shows the pfSense web interface for the System Update page. The breadcrumb trail is "System / Update / System Update". There are two tabs: "System Update" (active) and "Update Settings". A dark banner at the top of the main content area reads "Confirmation Required to update pfSense system.". Below this, there is a "Branch" dropdown menu set to "Latest stable version (2.5.2)". A note below the dropdown says: "Please select the branch from which to update the system firmware. Use of the development version is at your own risk!". Below the note is a table with two rows:

Current Base System	2.5.2
Latest Base System	2.5.2

At the bottom, there is a "Status" label followed by "Up to date." in green text.

Cliquer sur le logo PfSense pour revenir sur la page d'accueil.

Le routeur est désormais fonctionnel.

Déployer PfSense sur une machine virtuelle est une solution très simple pour connecter un réseau local virtualisé à Internet. De plus, pour faire évoluer l'installation, il donne la possibilité, à posteriori, de paramétrer un pare-feu et d'ouvrir d'autres segments réseau
